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(54) **HIGH ELECTRON AFFINITY DIELECTRIC LAYER TO IMPROVE CYCLING**

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(Continued)

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H10N 70/20 (2023.01)

G11C 13/00 (2006.01)

H10N 70/00 (2023.01)

(52) **U.S. Cl.**

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(2013.01); **H10N 70/021** (2023.02); **H10N 70/841** (2023.02)

(58) **Field of Classification Search**

CPC H10N 70/826; H10N 70/8833
See application file for complete search history.

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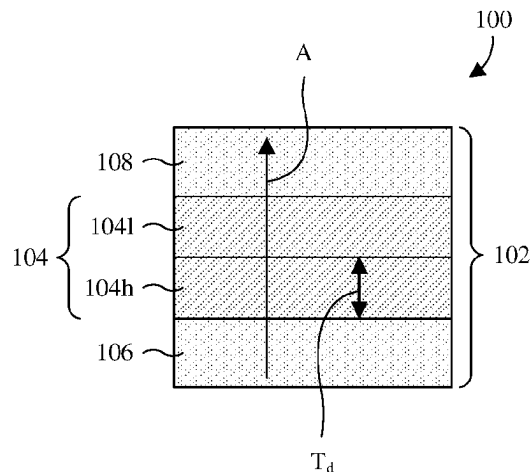
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(57)

ABSTRACT

Various embodiments of the present disclosure are directed towards a memory cell comprising a high electron affinity dielectric layer at a bottom electrode. The high electron affinity dielectric layer is one of multiple different dielectric layers vertically stacked between the bottom electrode and a top electrode overlying the bottom electrode. Further, the high electrode electron affinity dielectric layer has a highest electron affinity amongst the multiple different dielectric layers and is closest to the bottom electrode. The different dielectric layers are different in terms of material systems and/or material compositions. It has been appreciated that by arranging the high electron affinity dielectric layer closest to the bottom electrode, the likelihood of the memory cell

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becoming stuck during cycling is reduced at least when the memory cell is RRAM. Hence, the likelihood of a hard reset/failure bit is reduced.

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20 Claims, 16 Drawing Sheets

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(60) Provisional application No. 62/927,902, filed on Oct. 30, 2019.

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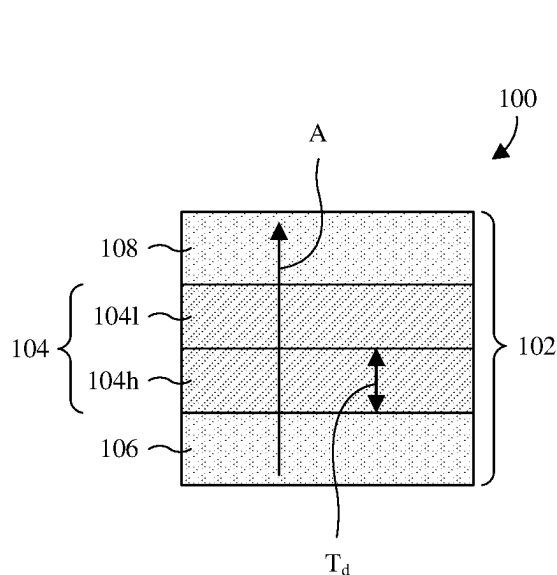
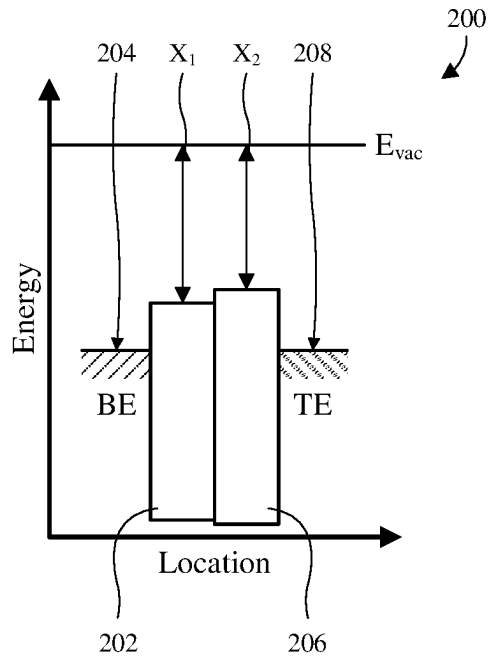
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**Fig. 1****Fig. 2**

Material	Electron Affinity
Silicon oxide (e.g., SiO_2)	0.55 eV
Silicon nitride (e.g., Si_3N_4)	1.65 eV
Tantalum oxide (e.g., Ta_2O_5)	3.75 eV
Barium titanium oxide (e.g., BaTiO_3)	4.05 eV
Barium zirconium oxide (e.g., BaZrO_3)	3.25 eV
Zirconium oxide (e.g., ZrO_2)	2.65 eV
Hafnium oxide (e.g., HfO_2)	2.55 eV
Lanthanum oxide (e.g., La_2O_3)	1.75 eV
Aluminum oxide (e.g., Al_2O_3)	1.25 eV
Yttrium oxide (e.g., Y_2O_3)	2.75 eV
Zirconium silicon oxide (ZrSiO_4)	2.55 eV
Titanium oxide (e.g., TiO_2)	2.85 eV

Fig. 3

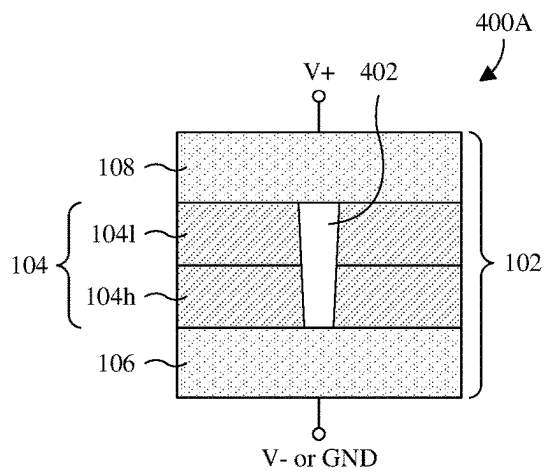


Fig. 4A

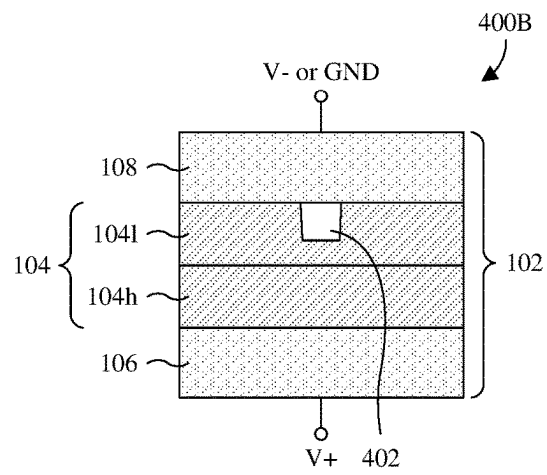


Fig. 4B

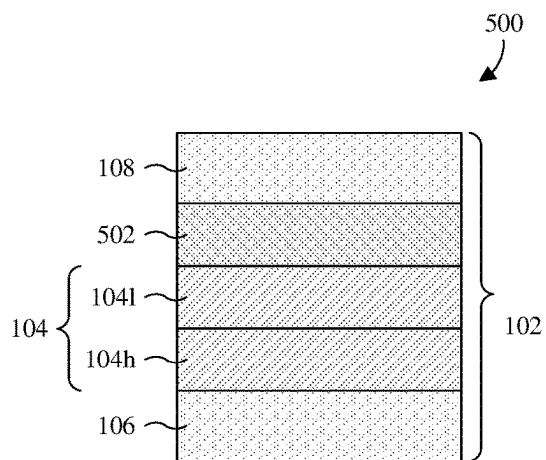


Fig. 5

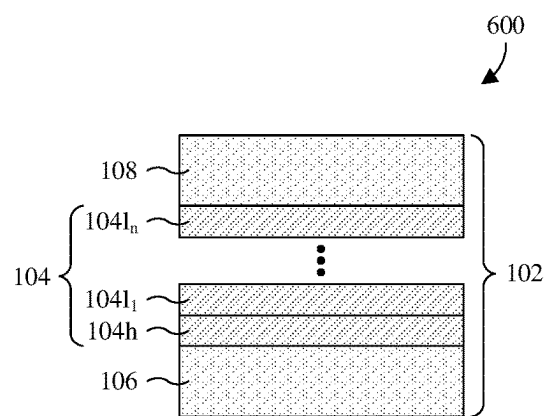


Fig. 6

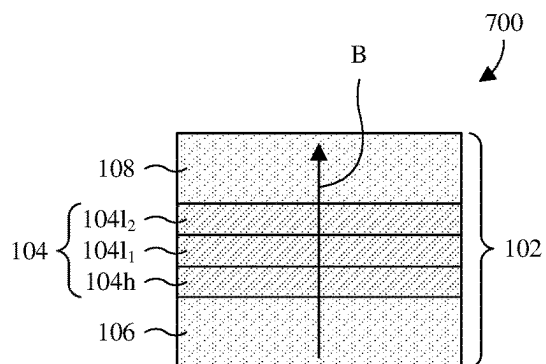


Fig. 7

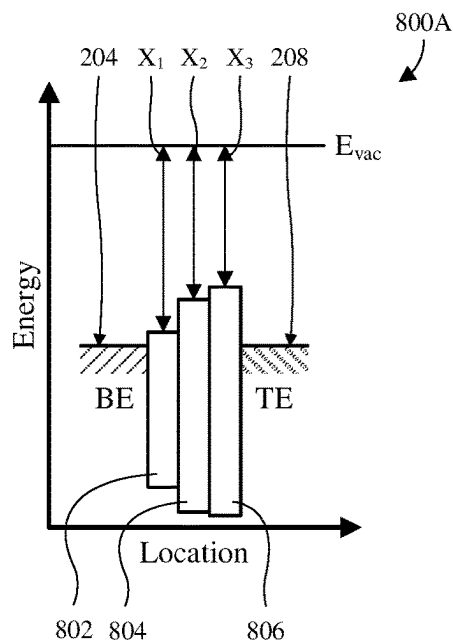


Fig. 8A

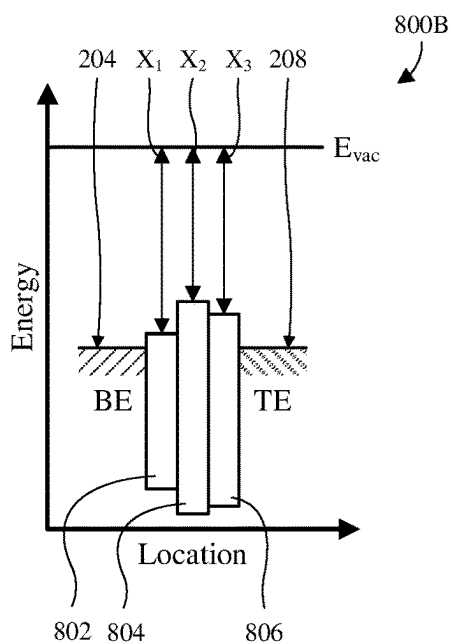


Fig. 8B

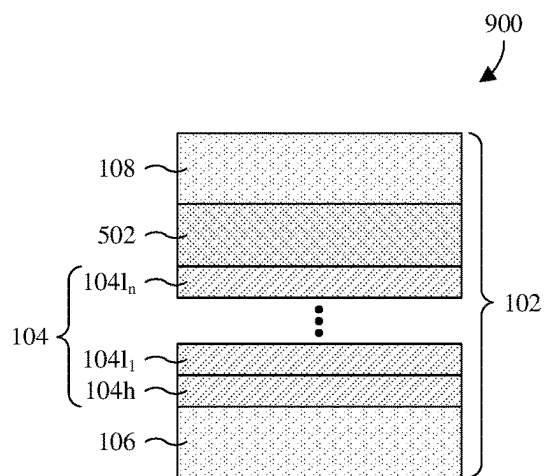


Fig. 9

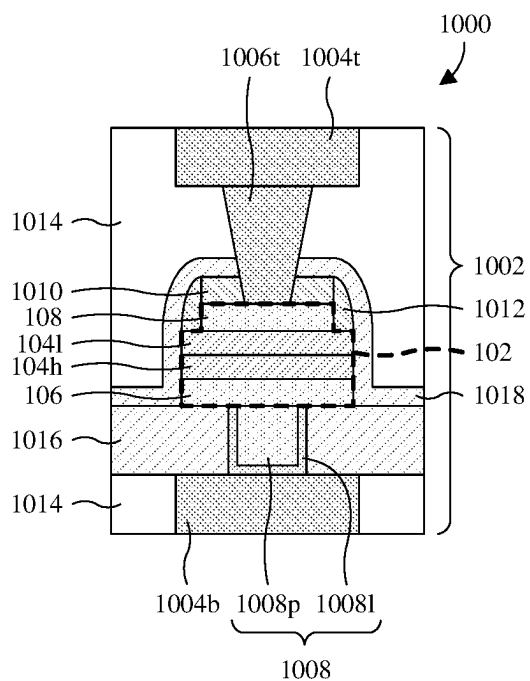


Fig. 10

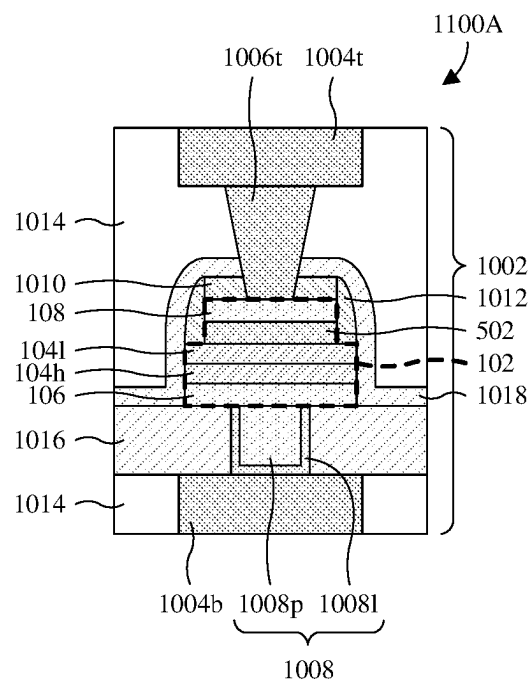


Fig. 11A

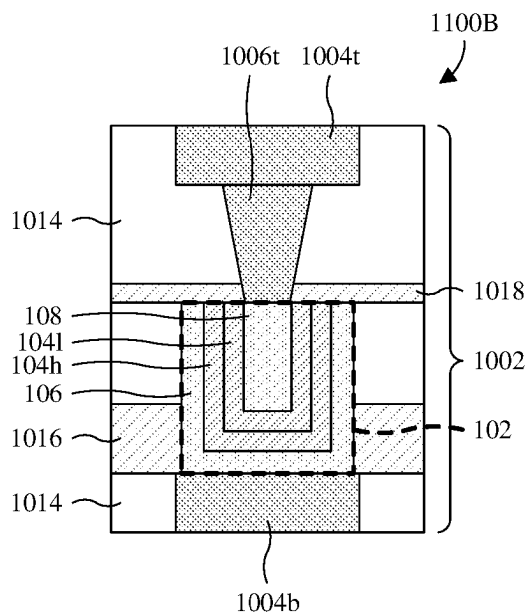


Fig. 11B

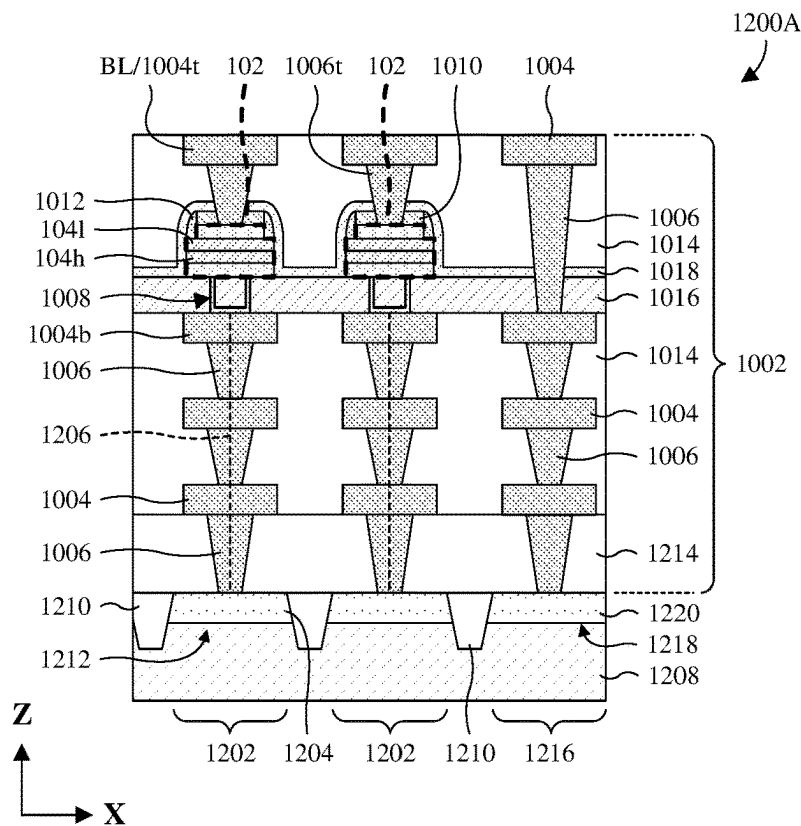


Fig. 12A

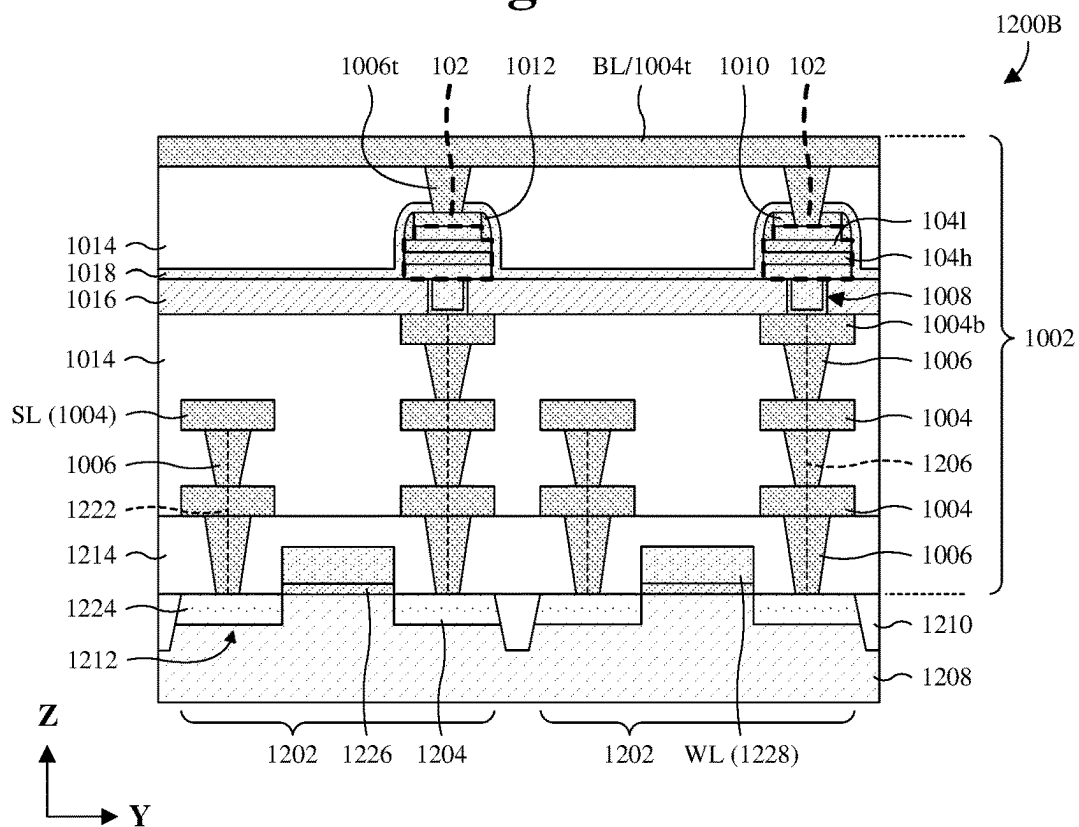


Fig. 12B

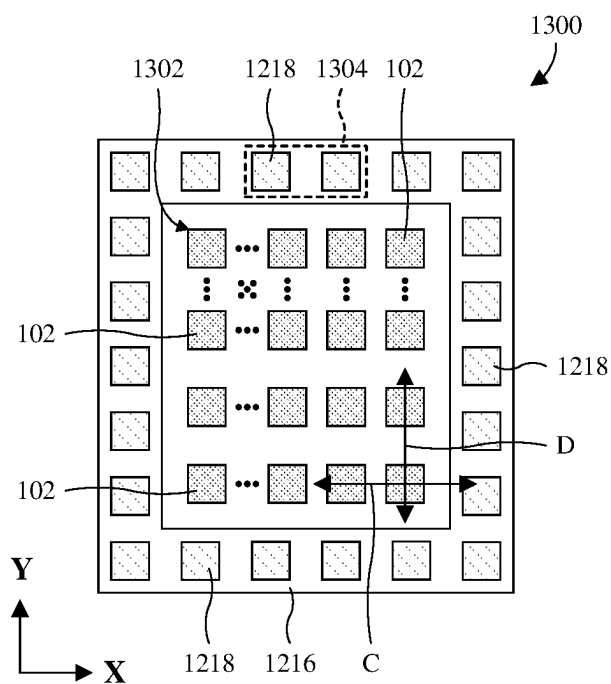


Fig. 13

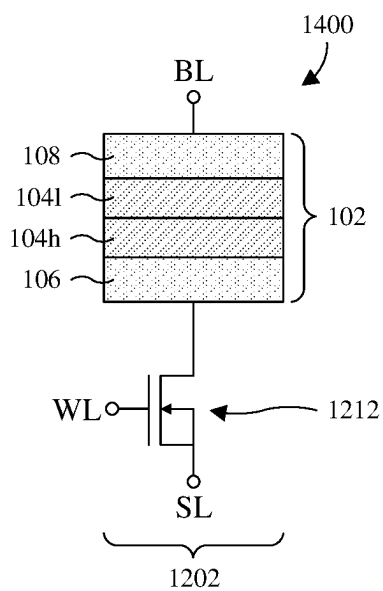


Fig. 14

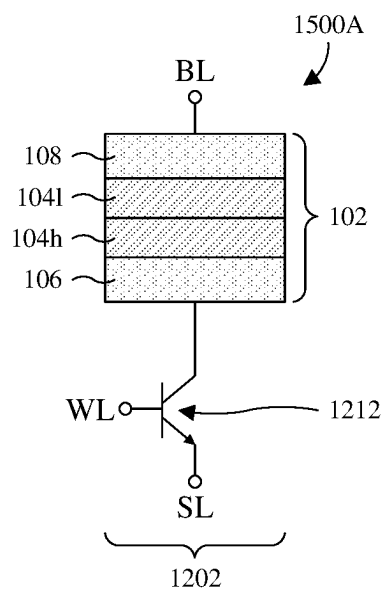


Fig. 15A

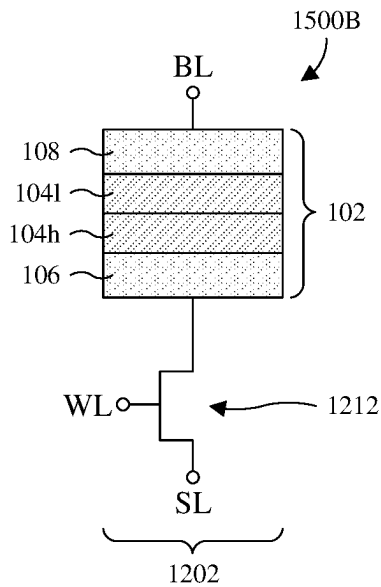


Fig. 15B

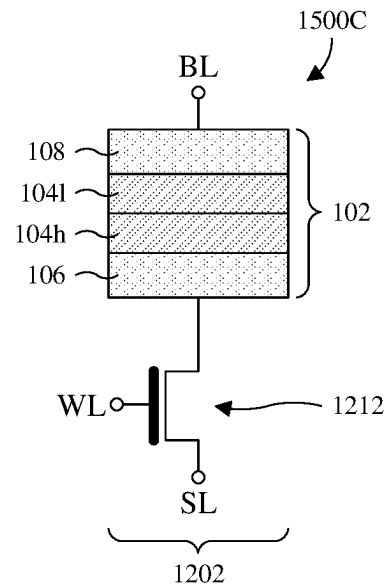


Fig. 15C

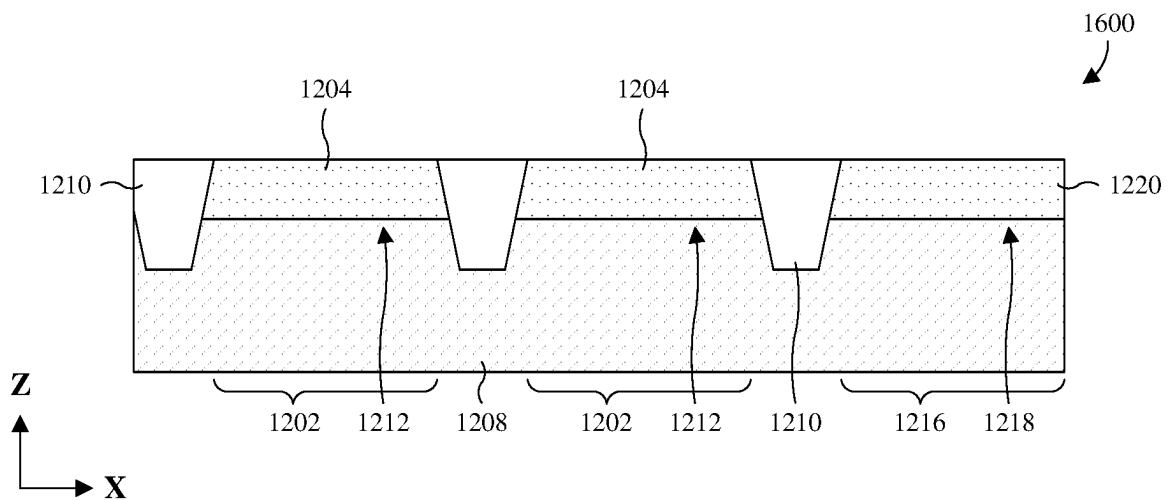


Fig. 16

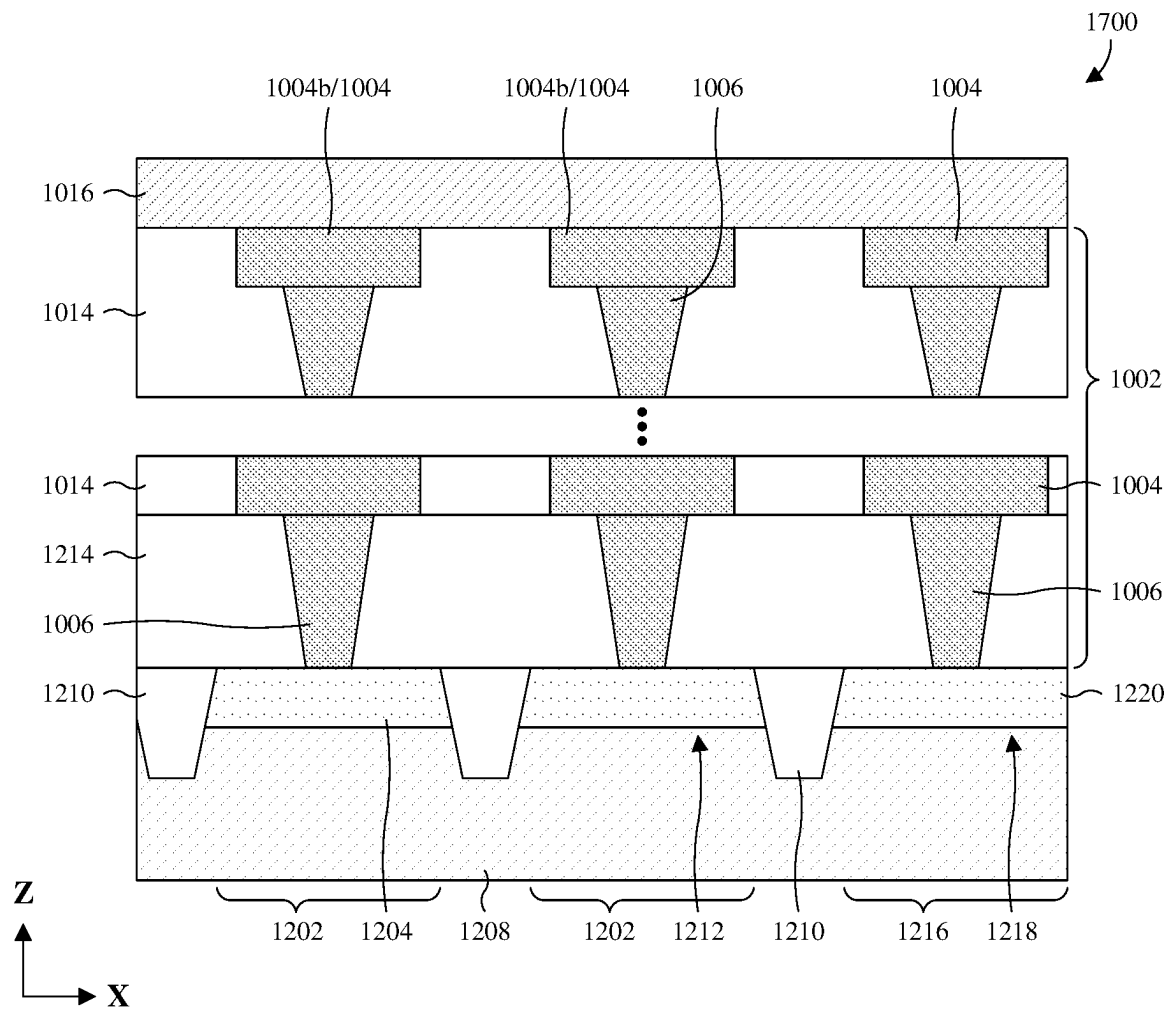


Fig. 17

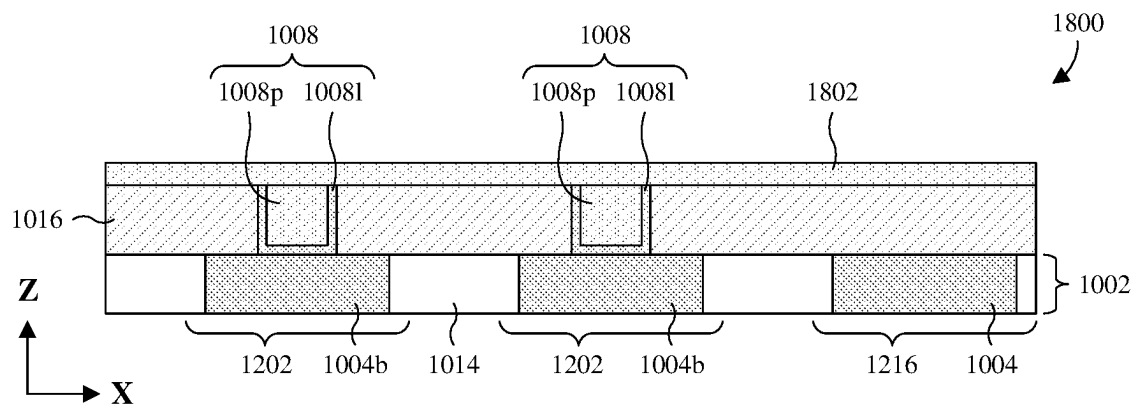


Fig. 18

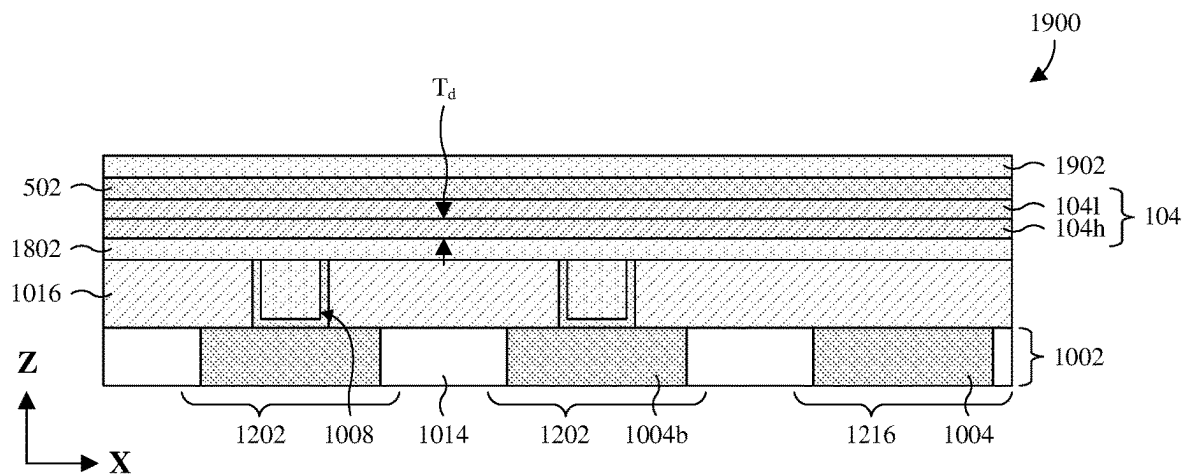


Fig. 19

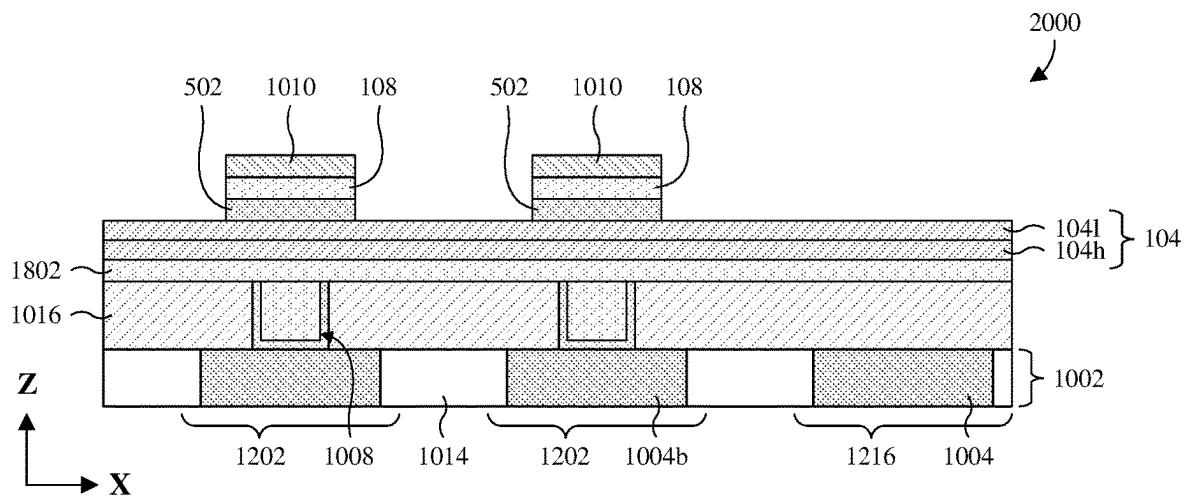
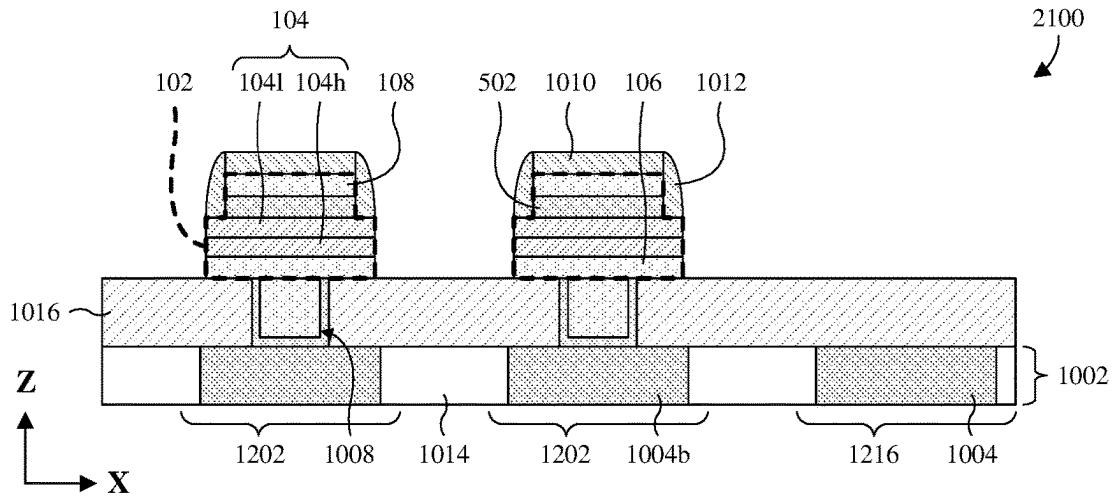
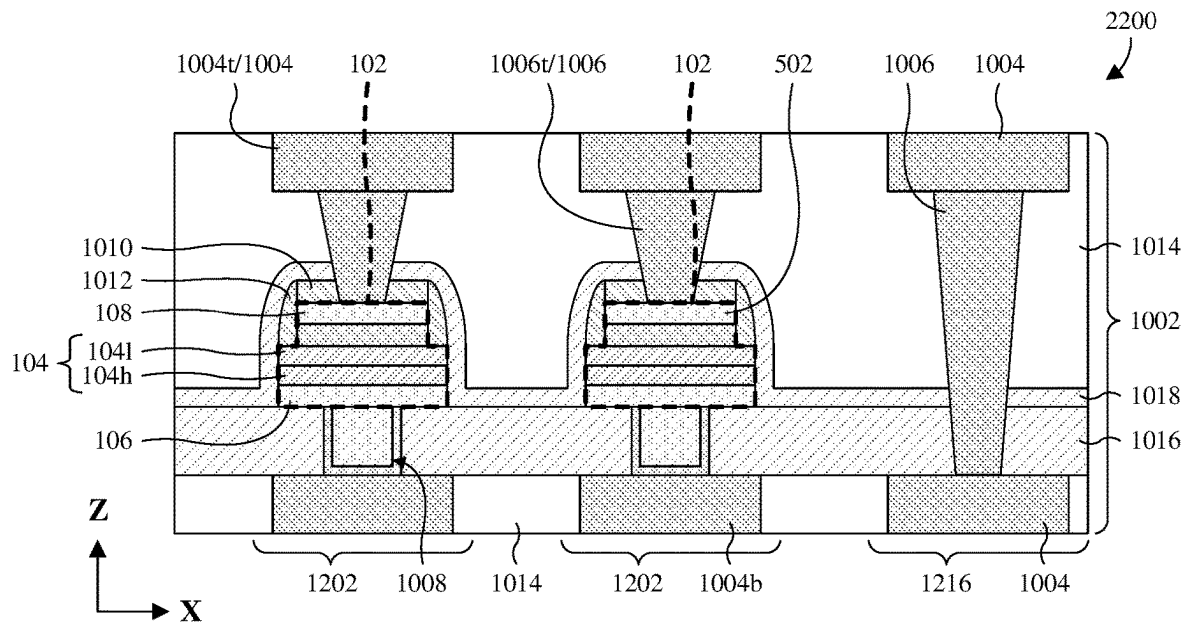


Fig. 20

**Fig. 21****Fig. 22**

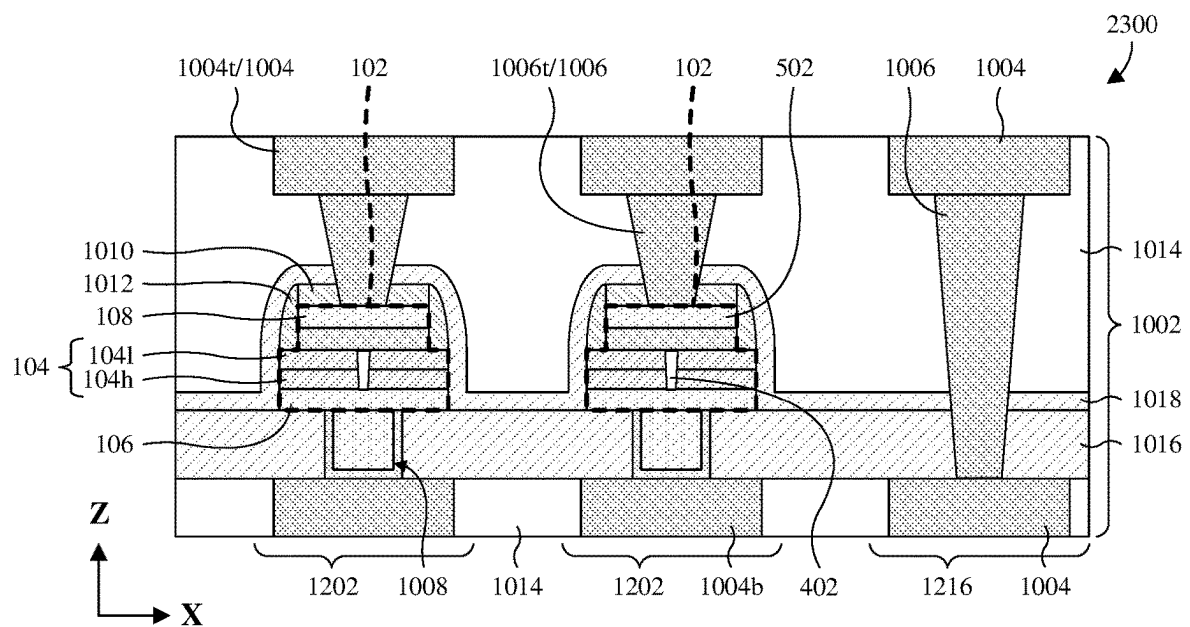
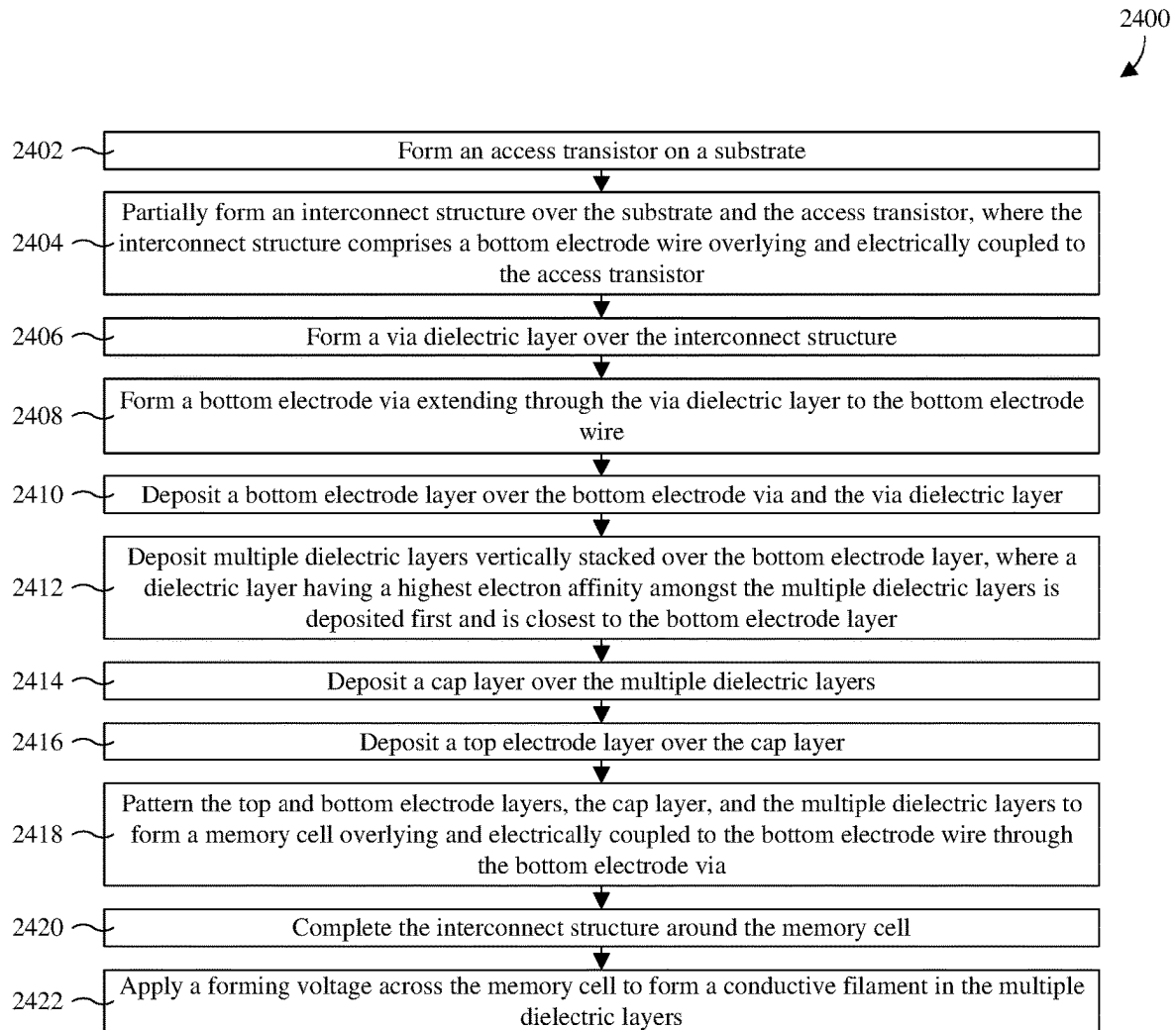


Fig. 23

**Fig. 24**

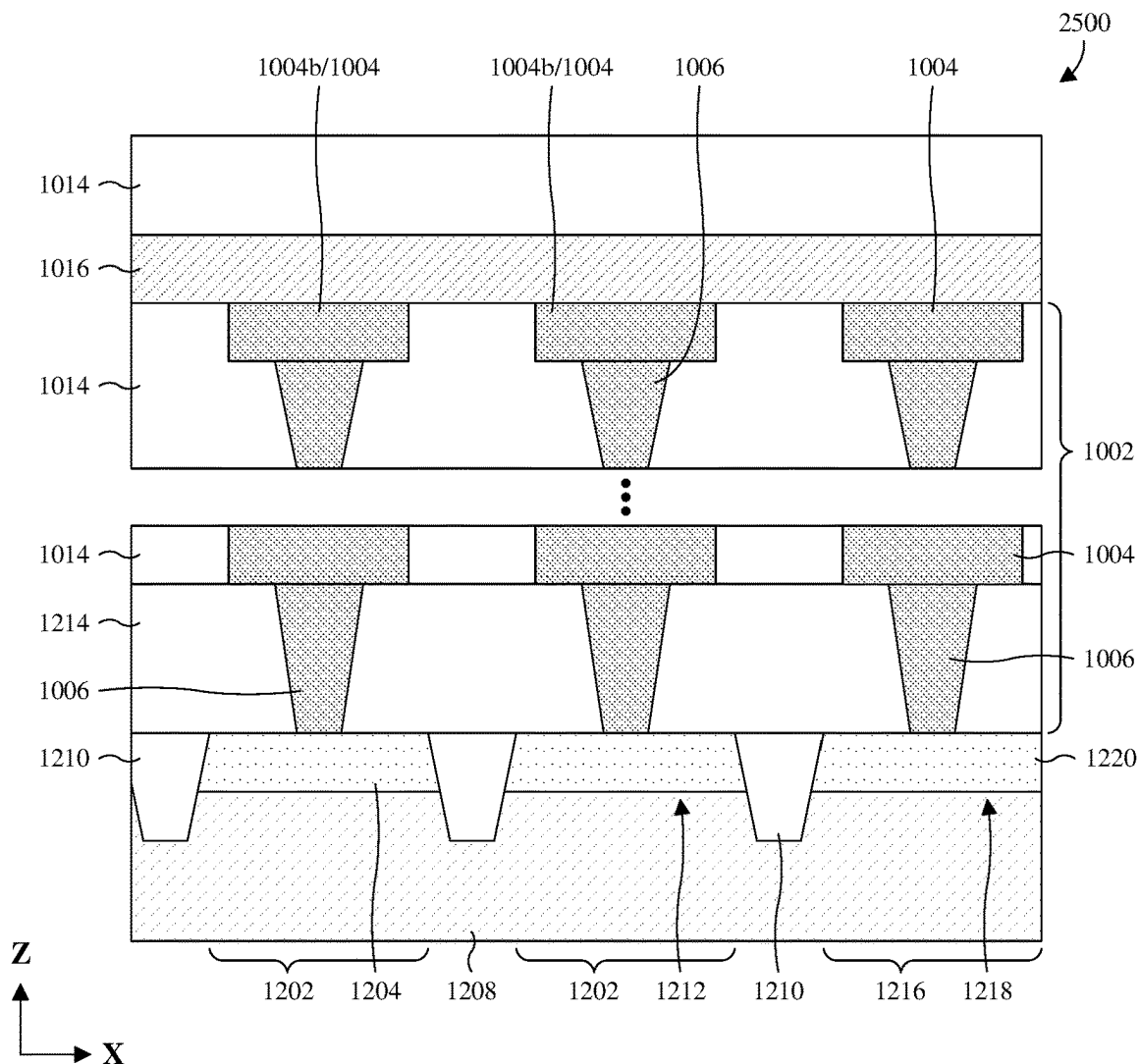


Fig. 25

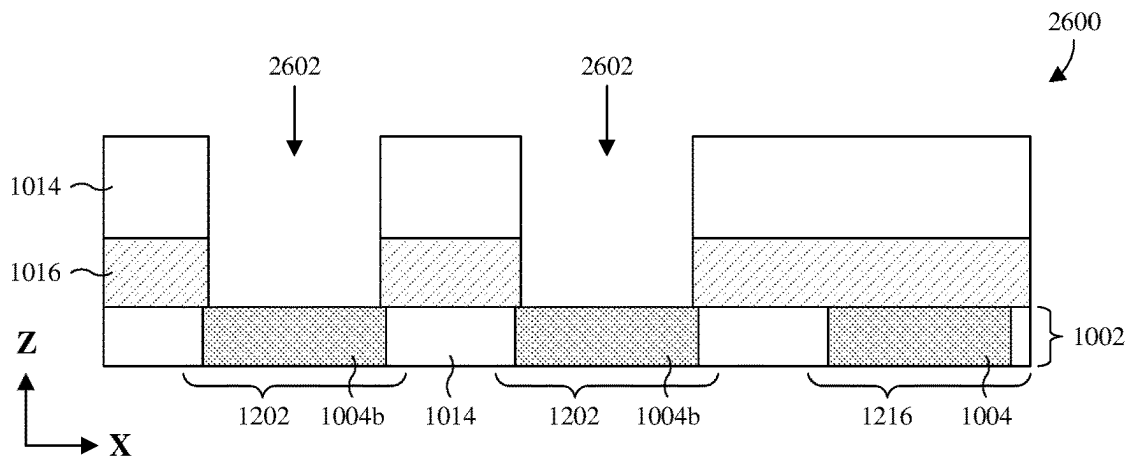


Fig. 26

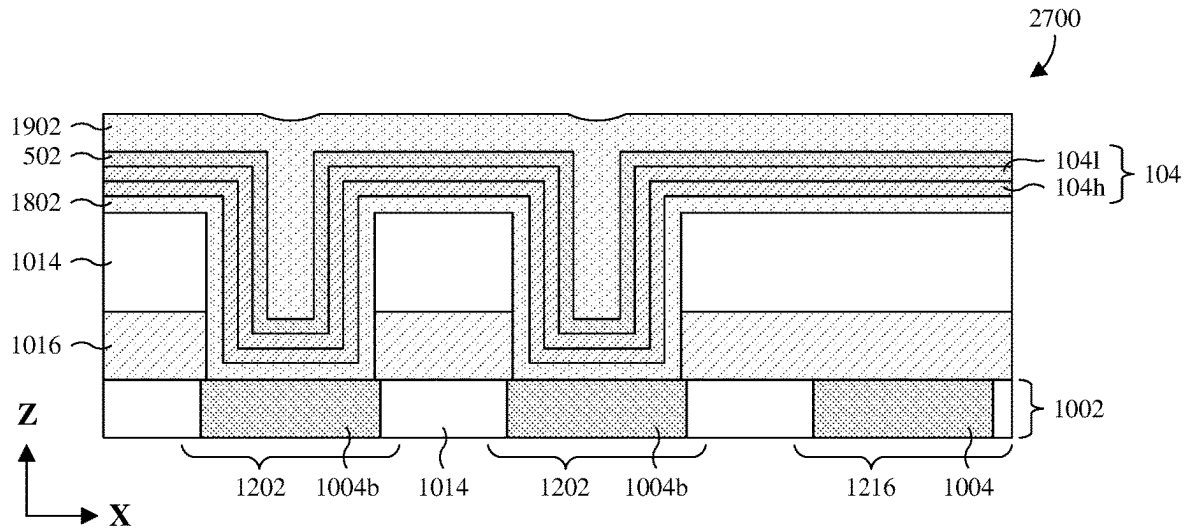


Fig. 27

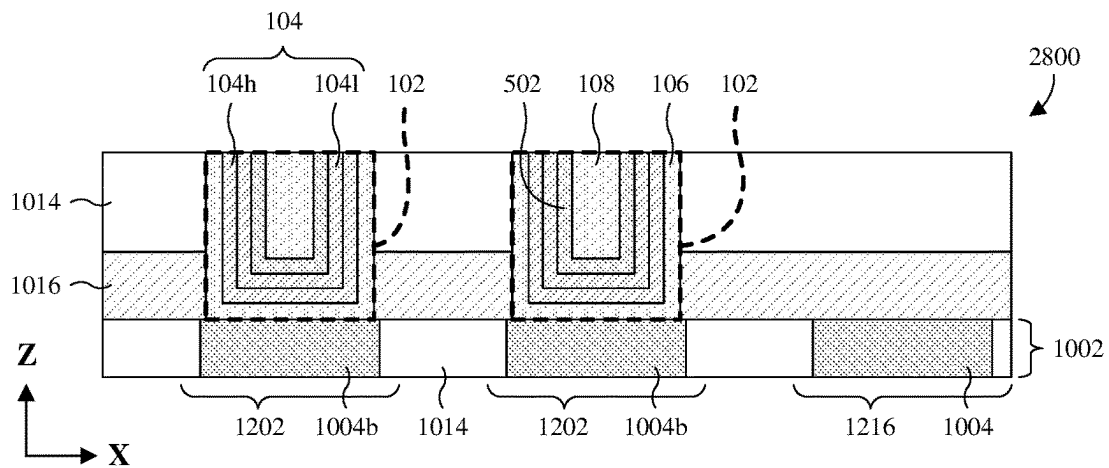


Fig. 28

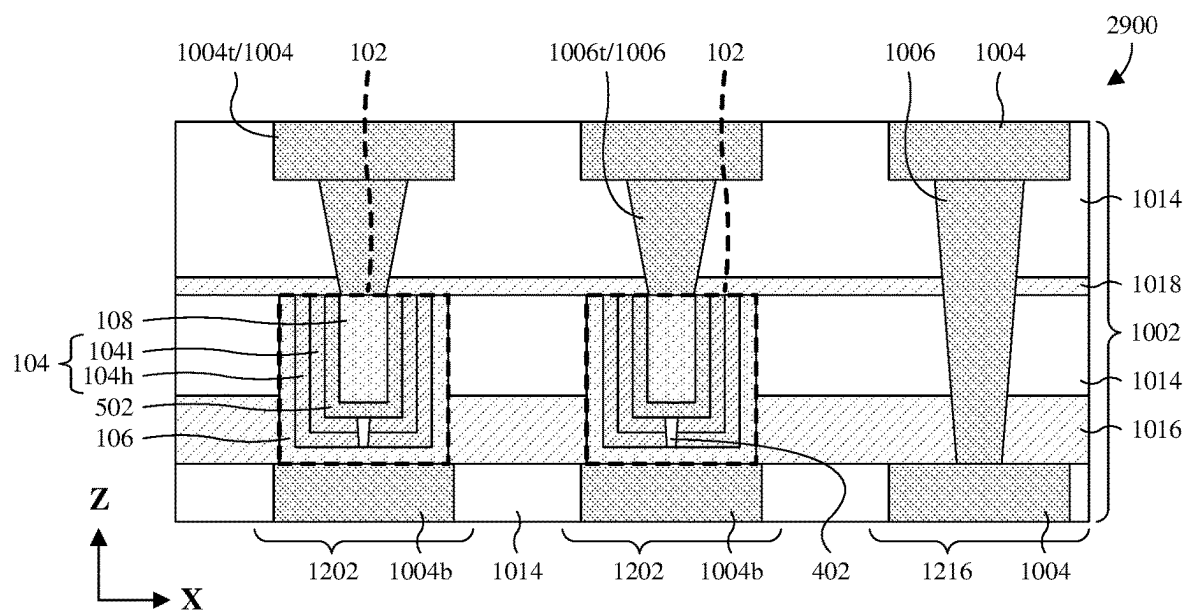
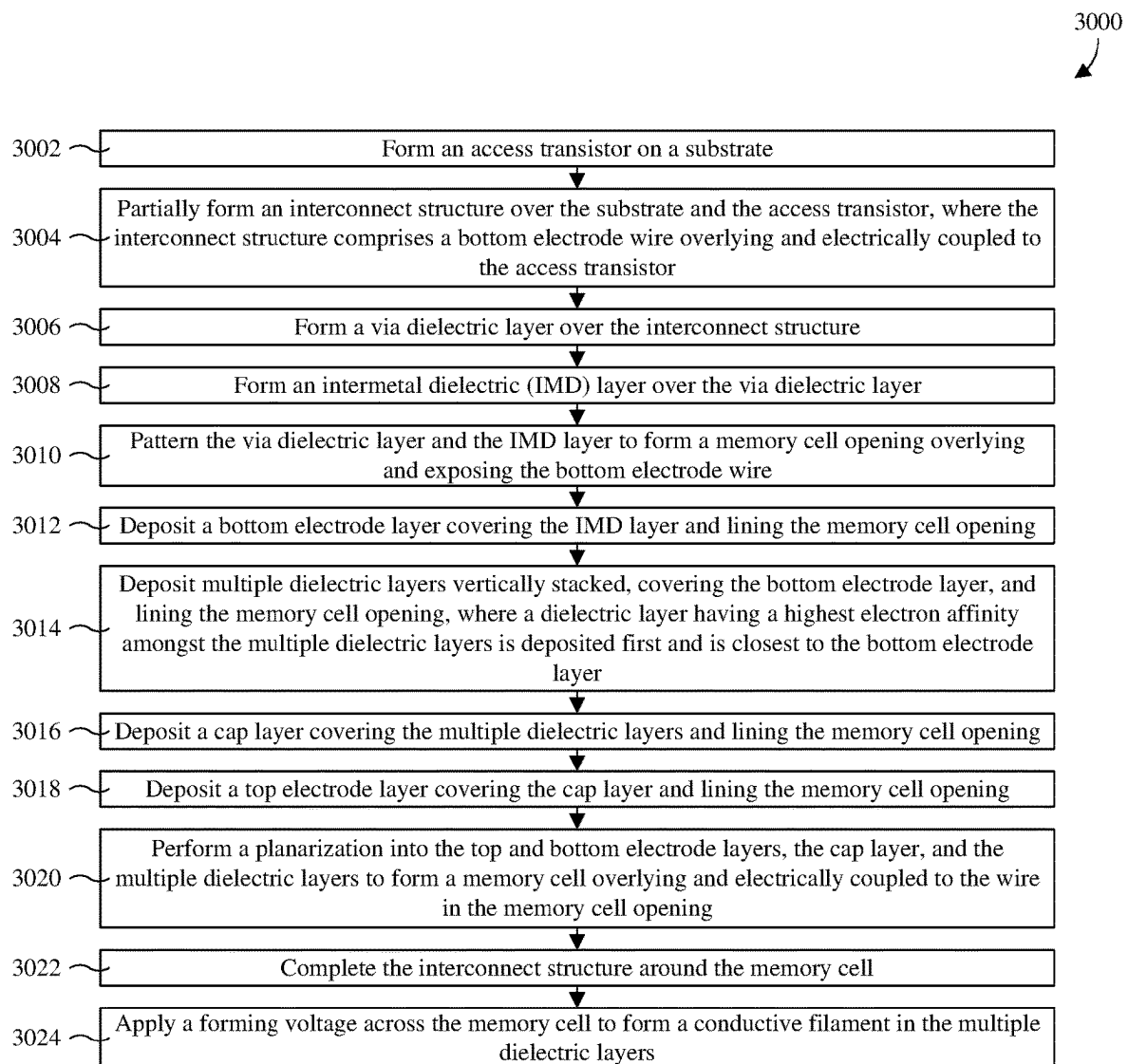


Fig. 29

**Fig. 30**

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HIGH ELECTRON AFFINITY DIELECTRIC LAYER TO IMPROVE CYCLING

REFERENCE TO RELATED APPLICATIONS

This Application is a Divisional of U.S. application Ser. No. 16/939,497, filed on Jul. 27, 2020, which claims the benefit of U.S. Provisional Application No. 62/927,902, filed on Oct. 30, 2019. The contents of the above-referenced Patent Applications are hereby incorporated by reference in their entirety.

BACKGROUND

Many modern-day electronic devices include non-volatile memory. Non-volatile memory is electronic memory that is able to store data in the absence of power. Some promising candidates for the next generation of non-volatile memory include oxygen-ion type resistive random-access memory (RRAM) and metal-ion type RRAM. Both types of RRAM have relatively simple structures and are compatible with complementary metal-oxide-semiconductor (CMOS) logic fabrication processes.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 illustrates a cross-sectional view of some embodiments of a memory cell in which a high electron affinity (HEA) dielectric layer is at a bottom electrode.

FIG. 2 illustrates a graph of some embodiments of electron affinities for dielectric layers, including the HEA dielectric layer, in the memory cell of FIG. 1.

FIG. 3 illustrates a table listing electron affinities for various materials.

FIGS. 4A and 4B illustrate cross-sectional views of some embodiments of the memory cell of FIG. 1 respectively during a set operation and a reset operation.

FIG. 5 illustrates a cross-sectional view of some alternative embodiments of the memory cell of FIG. 1 in which the memory cell further includes a cap layer.

FIG. 6 illustrates a cross-sectional view of some alternative embodiments of the memory cell of FIG. 1 in which the memory cell comprises three or more dielectric layers.

FIG. 7 illustrates a cross-sectional view of some embodiments of the memory cell of FIG. 6 in which the memory cell is limited to three dielectric layers.

FIGS. 8A and 8B illustrate graphs of different embodiments of electron affinities for dielectric layers, including the HEA dielectric layer, in the memory cell of FIG. 7.

FIG. 9 illustrates a cross-sectional view of some alternative embodiments of the memory cell of FIG. 6 in which the memory cell further includes a cap layer.

FIG. 10 illustrates a cross-sectional view of some embodiments of the memory cell of FIG. 1 in which the memory cell is integrated into an interconnect structure of an integrated circuit (IC) chip.

FIGS. 11A and 11B illustrate cross-sectional views of different alternative embodiments of the memory cell of FIG. 10.

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FIGS. 12A and 12B illustrate cross-sectional views of some embodiments of an IC chip comprising multiple memory cells integrated into individual one-transistor one-resistor (1T1R) cells and each configured as in FIG. 10.

FIG. 13 illustrates a top layout of some embodiments of the IC chip of FIGS. 12A and 12B.

FIG. 14 illustrates a schematic diagram of some embodiments of a 1T1R cell in FIGS. 12A and 12B.

FIGS. 15A-15C illustrate schematic diagrams of different alternative embodiments of the 1T1R cell of FIG. 14 in which access transistors are varied.

FIGS. 16-23 illustrate a series of cross-sectional views of some embodiments of a method for forming memory cells integrated with 1T1R cells and comprising individual HEA dielectric layers at corresponding bottom electrodes.

FIG. 24 illustrates a block diagram of some embodiments of the method of FIGS. 16-23.

FIGS. 25-29 illustrate a series of cross-sectional views of some alternative embodiments of the method of FIGS. 16-23 in which layers from which the memory cells are formed are patterned using a planarization.

FIG. 30 illustrates a block diagram of some embodiments of the method of FIGS. 25-29.

DETAILED DESCRIPTION

The present disclosure provides many different embodiments, or examples, for implementing different features of this disclosure. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

Some resistive random-access memory (RRAM) cells comprise a bottom electrode, a top electrode overlying the bottom electrode, and a single dielectric layer between the bottom and top electrodes. The single dielectric layer comprises a metal oxide and has a single material composition throughout. During a set operation, a set voltage with a positive polarity is applied from the top electrode to the bottom electrode to form a conductive filament in the single dielectric layer. The conductive filament electrically couples the bottom electrode to the top electrode, such that the RRAM cell is in a low resistance state (LRS). During a reset operation, a reset voltage with a negative polarity is applied from the top electrode to the bottom electrode to at least

partially dissolve the conductive filament. As such, the RRAM cell is in a high resistance state (HRS). Because a resistance of the RRAM cell changes during the set and reset operations, the resistance may be employed to represent a bit of data. For example, the LRS may represent a binary “1”, whereas the HRS may represent a binary “0”, or vice versa.

A challenge with the RRAM cell is that the RRAM cell may become stuck at the LRS when cycling the RRAM cell. To the extent that this occurs, it typically occurs after many cycles of the RRAM cell. While the RRAM cell is stuck, the RRAM cell may be known as a hard reset bit. Further, read current through the RRAM cell may be greater than or about the same as that when the RRAM cell is in the LRS. The RRAM cell typically remains stuck for many cycles and may hence be considered a hard failure bit. As such, the RRAM cell may be subject to error correction code (ECC) correction by an ECC device when the RRAM cell is one of many like RRAM cells defining an RRAM array. However, ECC devices may be designed to correct random soft failure bits, not hard failure bits, and hence hard failure bits may use ECC capacity intended for random soft failure bits. To the extent that hard failure bits use too much ECC capacity, an ECC device may become overwhelmed and may be unable to correct random soft failure bits. This may, in turn, lead to failure of an RRAM array.

Various embodiments of the present disclosure are directed towards a memory cell comprising a high electron affinity (HEA) dielectric layer at a bottom electrode, as well as a method for forming the memory cell. The memory cell may, for example, be an RRAM cell or some other suitable type of memory cell. The HEA dielectric layer is one of multiple different dielectric layers vertically stacked between the bottom electrode and a top electrode overlying the bottom electrode. Further, the HEA dielectric layer has a highest electron affinity, and is closest to the bottom electrode, amongst the different dielectric layers. The different dielectric layers are different in terms of material systems and/or material compositions. Different material systems correspond to the different sets of elements, whereas different material compositions correspond to different ratios of elements for the same set of elements.

It has been appreciated that by arranging the HEA dielectric layer closest to the bottom electrode, the likelihood of the memory cell becoming stuck during cycling is reduced at least when the memory cell is RRAM. Hence, the likelihood of a hard reset/failure bit is reduced. Because the likelihood of a hard reset/failure bit is reduced, ECC capacity for a memory array to which the memory cell belongs is less likely to be used by hard reset/failure bits. Because ECC capacity is less likely to be used by hard reset/failure bits, ECC capacity for the memory array is less likely to be overwhelmed by hard reset/failure bits. Hence, the likelihood of the memory array failing is less likely. Because the likelihood of the memory array failing is reduced by the HEA dielectric layer, no additional ECC capacity is needed. This, in turn, conserves integrated circuit (IC) chip area that may otherwise be employed for additional ECC capacity.

As will be seen hereafter, the HEA dielectric layer may be integrated into memory process flows with an additional deposition. As such, the HEA dielectric layer adds little to no extra cost during the memory process flows. Further, the integration is compatible with 40 nanometer process nodes and smaller.

With reference to FIG. 1, a cross-sectional view 100 of some embodiments of a memory cell 102 is provided in which a HEA dielectric layer 104h is at a bottom electrode 106. The memory cell 102 may, for example, be an oxygen-

ion type RRAM cell, a metal-ion type RRAM cell, or some other suitable type of memory cell. Note that metal-ion type RRAM cell may also be known as conductive bridging RAM (CBRAM).

The HEA dielectric layer 104h is one of multiple dielectric layers 104 that are vertically stacked over the bottom electrode 106 and that separate the bottom electrode 106 from a top electrode 108 overlying the bottom electrode 106. Further, the HEA dielectric layer 104h has a highest electron affinity, and is closest to the bottom electrode 106, amongst the multiple dielectric layers 104. Hence, the HEA dielectric layer 104h has a “high” electron affinity relative to a remainder of the multiple dielectric layers 104.

Electron affinity for a dielectric layer is an energy difference between a bottom conductive band edge of the dielectric layer and the vacuum level. The vacuum level is the same for the multiple dielectric layers 104 and may, for example, be about 4.05 electron volts (eV). Because the vacuum level is the same for the multiple dielectric layers 104, bottom conductive band edges of the multiple dielectric layers 104 respectively define electron affinities of the multiple dielectric layers 104. Further, the bottom conductive band edge of the HEA dielectric layer 104h is the lowest amongst the multiple dielectric layers 104.

It has been appreciated that by arranging the HEA dielectric layer 104h closest to the bottom electrode 106, the likelihood of the memory cell 102 becoming stuck during cycling is reduced at least when the memory cell 102 is RRAM. For example, the memory cell 102 may be most likely to become stuck during a reset operation when transitioning from a LRS to a HRS. Because the HEA dielectric layer 104h has the highest electron affinity amongst the multiple dielectric layers 104 and is at the bottom electrode 106, electrons may more easily pass through the multiple dielectric layers 104 during the reset operation. Hence, the likelihood of the memory cell 102 becoming stuck during cycling is reduced.

Because the likelihood of the memory cell 102 becoming stuck during cycling is reduced, the likelihood of a hard reset/failure bit is reduced. Because the likelihood of a hard reset/failure bit is reduced, ECC capacity for a memory array (not shown) incorporating the memory cell 102 is less likely to be used by hard reset/failure bits. Hence, the likelihood of the memory array failing is less likely. Because the likelihood of the memory array failing is reduced by the HEA dielectric layer 104h, no additional ECC capacity and therefore no additional IC chip area are needed.

With continued reference to FIG. 1, the multiple dielectric layers 104 are vertically stacked, and define a dielectric structure extending, from the bottom electrode 106 to the top electrode 108. Further, the multiple dielectric layers 104 have two dielectric layers: 1) the HEA dielectric layer 104h; and 2) a low electron affinity (LEA) dielectric layer 104l. The LEA dielectric layer 104l overlies and neighbors the HEA dielectric layer 104h and has a “low” electron affinity relative to the HEA dielectric layer 104h. In alternative embodiments, the multiple dielectric layers 104 have three or more different dielectric layers.

Each of the HEA and LEA dielectric layers 104h, 104l may, for example, be a metal oxide, a metal oxynitride, a component metal oxide, some other suitable dielectric(s), or any combination of the foregoing. Further, each of the HEA and LEA dielectric layers 104h, 104l may, for example, be titanium oxide (e.g., TiO₂), hafnium oxide (e.g., HfO₂), hafnium aluminum oxide (e.g., Hf_xAl_{1-x}O₂), tantalum oxide (e.g., Ta₂O₅), hafnium tantalum oxide (e.g., Hf_xTa_{1-x}O₂), tungsten oxide (e.g., WO₂), zirconium oxide (e.g., ZrO₂),

aluminum oxide (e.g., Al_2O_3), sulfated tin oxide (e.g., STO), some other suitable dielectric(s), or any combination of the foregoing. In some embodiments, the HEA dielectric layer **104h** is a high k dielectric and/or the LEA dielectric layer **104l** is a high k dielectric. A high k dielectric may, for example, be a dielectric with a dielectric constant k greater than about 3.9, 10.0, or some other suitable value.

The HEA and LEA dielectric layers **104h**, **104l** have different material systems or different material compositions. Different material systems correspond to different sets of elements. Different material compositions correspond to different ratios of elements for the same set of elements (e.g., the same material systems).

In embodiments in which the HEA and LEA dielectric layers **104h**, **104l** have different material systems, the HEA dielectric layer **104h** consists of or consists essentially of a first set of elements and the LEA dielectric layer **104l** consists of or consists essentially of a second set of elements that is different than the first set of elements. For example, the HEA dielectric layer **104h** may be aluminum oxide (e.g., Al_2O_3) and the LEA dielectric layer **104l** may be silicon oxide (e.g., SiO_2). Other suitable materials are, however, amenable.

In embodiments in which the HEA and LEA dielectric layers **104h**, **104l** have different material compositions, the HEA dielectric layer **104h** consists of or consists essentially of a set of elements. Further, the LEA dielectric layer **104l** consists of or consists essentially of the set of elements but has a different ratio of the elements compared to the HEA dielectric layer **104h**. For example, the HEA dielectric layer **104h** may be aluminum oxide (e.g., Al_2O_3) and the LEA dielectric layer **104l** may be aluminum oxide with a different ratio of aluminum and oxide (e.g., Al_xO_y , where $x \neq 2$ and $y \neq 3$). Other suitable materials are, however, amenable.

The HEA and LEA dielectric layers **104h**, **104l** have individual thicknesses T_d . In some embodiments, the thicknesses T_d are about 1-50 nanometers, about 1-25 nanometers, about 25-50 nanometers, or some other suitable value. If the thickness T_d of the HEA or LEA dielectric layer **104h**, **104l** is too small (e.g., less than about 1 nanometer or some other suitable value), benefits from material properties of the dielectric layer may not be attained. For example, if the thickness T_d of the HEA dielectric layer **104h** is too small, the HEA dielectric layer **104h** may not reduce the likelihood of the memory cell **102** becoming stuck. If the thickness T_d of the HEA or LEA dielectric layer **104h**, **104l** is too large (e.g., more than about 50 nanometers or some other suitable value), operating voltages of the memory cell **102** may be too high. The high voltages may, for example, increase power consumption, reduce the lifespan of the memory cell **102**, and increase the risk of device failure.

The top and bottom electrodes **106**, **108** are conductive and border the multiple dielectric layers **104**. The bottom electrode **106** may, for example, be or comprise a metal, a metal nitride, a metal oxide, doped polysilicon, some other suitable conductive material(s), or any combination of the foregoing. The top electrode **108** may, for example, be or comprise a metal, a metal nitride, doped polysilicon, some other suitable conductive material(s), or any combination of the foregoing. The top and bottom electrodes **106**, **108** may, for example, each be or comprise aluminum (e.g., Al), titanium (e.g., Ti), tantalum (e.g., Ta), gold (e.g., Au), platinum (e.g., Pt), tungsten (e.g., W), nickel (e.g., Ni), iridium (e.g., Ir), titanium nitride (e.g., TiN), tantalum nitride (e.g., TaN), N-doped polysilicon, P-doped polysilicon, some other suitable conductive material(s), or any combination of the foregoing.

With reference to FIG. 2, a graph **200** of some embodiments of electron affinities for the HEA and LEA dielectric layers **104h**, **104l** of FIG. 1 is provided. The vertical axis corresponds to energy and the horizontal axis corresponds to position along line A in FIG. 1. A first band gap **202** of the HEA dielectric layer **104h** is spaced from a vacuum energy level E_{vac} by a first electron affinity X_1 and borders a first fermi level **204** of the bottom electrode **106**. A second band gap **206** of the LEA dielectric layer **104l** is spaced from the vacuum energy level E_{vac} by a second electron affinity X_2 less than the first electron affinity X_1 . Further, the second band gap **206** borders a second fermi level **208** of the top electrode **108**. The first and second fermi levels **204**, **208** are the same, but may alternatively be different.

Note that a top edge of the first band gap **202** corresponds to a bottom conductive band edge of the HEA dielectric layer **104h**, whereas a top edge of the second band gap **206** corresponds to a bottom conductive band edge of the LEA dielectric layer **104l**. Hence, the bottom conductive band edge of the HEA dielectric layer **104h** is lower than the bottom conductive band edge of the LEA dielectric layer **104l**.

As described above, it has been appreciated that arranging the HEA dielectric layer **104h** closest to the bottom electrode **106** reduces the likelihood of the memory cell **102** becoming stuck during cycling at least when the memory cell is RRAM. Hence, the likelihood of a hard reset/failure bit is reduced. Because the likelihood of a hard reset/failure bit is reduced, ECC capacity for a memory array (not shown) incorporating the memory cell **102** is less likely to be used by hard reset/failure bits. Hence, the likelihood of the memory array failing is less likely.

To determine the first and second electron affinities X_1 , X_2 respectively of the HEA and LEA dielectric layers **104h**, **104l**, the first and second electron affinities X_1 , X_2 may be measured by x-ray photoelectron spectroscopy (XPS) or by other suitable methods. Alternatively, materials of the HEA and LEA dielectric layers **104h**, **104l** may be looked up in the table of FIG. 3, which lists materials and corresponding electron affinities. For example, supposing the HEA dielectric layer **104h** is Al_2O_3 and the LEA dielectric layer **104l** is SiO_2 , it can be seen through reference to the table of FIG. 3 that the first and second electron affinities X_1 , X_2 are respectively about 1.25 eV and about 0.55 eV.

If a material of the HEA or LEA dielectric layer **104h**, **104l** is not found in the table of FIG. 3, but is a mixture of two or more materials found in the table of FIG. 3, an electron affinity of the material may be calculated from electron affinities of the two or more materials. Particularly, the electron affinity of the material may be calculated as a weighted summation of the electron affinities of the two or more materials using atomic percentages of the two or more materials as weights respectively for the electron affinities. For example, suppose the HEA or LEA dielectric layer **104h**, **104l** is ZrTiO_4 . ZrTiO_4 is not found in the table of FIG. 3 but is a mixture of TiO_2 and ZrO_2 . TiO_2 and ZrO_2 each have an atomic percentage of about 50 in ZrTiO_4 . Further, as seen through reference to the table of FIG. 3, electron affinities for ZrO_2 and TiO_2 are respectively about 2.65 eV and 2.85 eV. Therefore, an electron affinity of ZrTiO_4 is about equal to $2.65 \text{ eV} \cdot 0.5 + 2.85 \text{ eV} \cdot 0.5$, which is about equal to 2.75 eV. Notwithstanding that the determination of electron affinity was illustrated using ZrTiO_4 , other suitable materials are amenable.

With reference to FIGS. 4A and 4B, cross-sectional views **400A**, **400B** of some embodiments of the memory cell **102** of FIG. 1 respectively during a set operation and a reset

operation are provided. At FIG. 4A, a set voltage (e.g., V+ to V- or to GND) with a positive polarity is applied from the top electrode 108 to the bottom electrode 106 to form a conductive filament 402 in the multiple dielectric layers 104. The conductive filament 402 electrically couples the top electrode 108 to the bottom electrode 106 so the memory cell 102 is in a LRS. At FIG. 4B, a reset voltage (e.g., V- to V+ or GND to V+) with a negative polarity is applied from the top electrode 108 to the bottom electrode 106 to at least partially dissolve the conductive filament 402. Because the conductive filament 402 is at least partially dissolved, the memory cell 102 is in a HRS. Because a resistance of the memory cell 102 changes during the set and reset operations, the resistance may be employed to represent a bit of data. For example, the LRS may represent a binary "1", whereas the HRS may represent a binary "0", or vice versa.

In some embodiments, the memory cell 102 is an oxygen-ion type RRAM cell. During the set operation for at least some embodiments of the oxygen-ion type RRAM cell, oxygen ions move from the multiple dielectric layers 104 to a reservoir region (not shown) between the multiple dielectric layers 104 and the top electrode 108. Movement of the oxygen ions leaves oxygen vacancies making up the conductive filament 402 in the multiple dielectric layers 104. During the reset operation for at least some embodiments of the oxygen-ion type RRAM cell, oxygen ions move from the reservoir region to the multiple dielectric layers 104 to fill the oxygen vacancies and to break the conductive filament 402.

In some embodiments, the memory cell 102 is a metal-ion type RRAM cell. During the set operation for at least some embodiments of the metal-ion type RRAM cell, the top electrode 108 oxidizes to form metal ions. Further, the metal ions migrate to the multiple dielectric layers 104 and reduce into the conductive filament 402. During the reset operation for at least some embodiments of the metal-ion type RRAM cell, the conductive filament 402 oxidizes to form metal ions. Further, the metal ions migrate to the top electrode 108 and reduce into the top electrode 108. This, in turn, breaks the conductive filament 402. In some embodiments in which the memory cell 102 is the metal-ion type RRAM cell, the conductive filament 402 is flipped vertically in FIGS. 4A and 4B and extends upward from the bottom electrode 106 in FIGS. 4A and 4B.

With reference to FIG. 5, a cross-sectional view 500 of some alternative embodiments of the memory cell 102 of FIG. 1 is provided in which the memory cell 102 further comprises a cap layer 502. The cap layer 502 is between the top electrode 108 and the multiple dielectric layers 104 and has a high affinity for oxygen compared to the bottom and top electrodes 106, 108. In other words, the cap layer 502 depends upon less energy to react with oxygen than the bottom and top electrodes 106, 108.

In at least some embodiments in which the memory cell 102 is an oxygen-ion type RRAM cell, the reservoir region (not shown) is between the multiple dielectric layers 104 and the cap layer 502. Further, because the cap layer 502 has the high affinity for oxygen, the cap layer 502 increases the size of the reservoir region. As such, more oxygen ions may be stored while the memory cell 102 is set and hence in the LRS. Because more oxygen ions may be stored, the conductive filament 402 (see, e.g., FIGS. 4A and 4B) may have a larger density of oxygen vacancies and hence the LRS may have a lower resistance. This, in turn, may increase the difference between the resistances of the memory cell 102 respectively in the LRS and the HRS and may hence increase the switching window.

In at least some embodiments in which the memory cell 102 is a metal-ion type RRAM cell, the cap layer 502 oxidizes instead of the top electrode 108 during the set operation. Further, because the cap layer 502 has the high affinity for oxygen relative to the top electrode 108, the cap layer 502 more readily oxidizes than the top electrode 108. As such, the set operation may use a smaller set voltage, which may improve power efficiency and/or may increase the lifespan of the memory cell 102. Further, the density of metal ions may be higher and hence the density of metal in the conductive filament 402 may be higher. This, in turn, may increase the difference between the resistances of the memory cell 102 respectively in the LRS and the HRS and may hence increase the switching window.

The cap layer 502 may, for example, be or comprise aluminum, titanium, tantalum, hafnium, titanium oxide, hafnium oxide, zirconium oxide, germanium oxide, cerium oxide, some other suitable material(s), or any combination of the foregoing. In some embodiments, the cap layer 502 is conductive and/or is metal. For example, in embodiments in which the memory cell 102 is a metal-ion type RRAM cell, the cap layer 502 is conductive and comprises metal. In alternative embodiments, the cap layer 502 is dielectric. In embodiments in which the cap layer 502 is dielectric, the cap layer 502 has a lower electron affinity and a higher bottom conductive band edge than the HEA dielectric layer 104h.

With reference to FIG. 6, a cross-sectional view 600 of some alternative embodiments of the memory cell 102 of FIG. 1 is provided in which the multiple dielectric layers 104 include three or more dielectric layers: 1) the HEA dielectric layer 104h; and 2) two or more LEA dielectric layers (labeled 104l₁ to 104l_n, where n is an integer greater than 1).

Each of the two or more LEA dielectric layers 104l₁, . . . , 104l_n is as the LEA dielectric layer 104l of FIG. 1 is described. Hence, each of the two or more LEA dielectric layers 104l₁, . . . , 104l_n has a lower electron affinity than the HEA dielectric layer 104h. Further, each of the two or more LEA dielectric layers 104l₁, . . . , 104l_n has a different material system or a different material composition than the HEA dielectric layer 104h. In some embodiments, each of the two or more LEA dielectric layers 104l₁, . . . , 104l_n is different than each neighboring LEA dielectric layer and/or each other LEA dielectric layer.

With reference to FIG. 7, a cross-sectional view 700 of some embodiments of the memory cell 102 of FIG. 6 is provided in which the memory cell 102 is limited to two LEA dielectric layers: a first LEA dielectric layer 104l₁; and a second LEA dielectric layer 104l₂. In other words, the integer n in FIG. 6 is equal to 2.

With reference to FIGS. 8A and 8B, graphs 800A and 800B of different embodiments of electron affinities for the HEA dielectric layer 104h of FIG. 7 and the first and second and second LEA dielectric layers 104l₁, 104l₂ of FIG. 7 are provided. The vertical axis corresponds to energy and the horizontal axis corresponds to position along line B in FIG. 7.

A first band gap 802 of the HEA dielectric layer 104h is spaced from a vacuum energy level E_{vac} by a first electron affinity X₁ and borders a first fermi level 204 of the bottom electrode 106. A second band gap 804 of the first LEA dielectric layer 104l₁ is spaced from the vacuum energy level E_{vac} by a second electron affinity X₂. A third band gap 806 of the second LEA dielectric layer 104l₂ is spaced from the vacuum energy level E_{vac} by a third electron affinity X₃ and borders a second fermi level 208 of the top electrode 108. As explained above, the second and third electron affinities X₂, X₃ are less than the first electron affinity X₁ to prevent the

memory cell **102** from becoming stuck during cycling at least when the memory cell is RRAM.

Note that a top edge of the first band gap **802** corresponds to a bottom conductive band edge of the HEA dielectric layer **104h**, and top edges of the second and third band gaps **804**, **806** correspond to bottom conductive band edges of the first and second LEA dielectric layers **104l₁**, **104l₂**. Hence, the bottom conductive band edge of the HEA dielectric layer **104h** is lower than the bottom conductive band edges of the first and second LEA dielectric layers **104l₁**, **104l₂**.

With reference specifically to FIG. **8A**, the first electron affinity X_1 is greater than the second electron affinity X_2 , which is greater than the third electron affinity X_3 . Hence, the electron affinity of a dielectric structure defined by the multiple dielectric layers **104** of FIG. **7** discretely decreases from the bottom electrode **106** to the top electrode **108**. Further, the bottom conductive band edge of the dielectric structure has a stepped profile stepping upward from the bottom electrode **106** to the top electrode **108**.

With reference specifically to FIG. **8B**, the first electron affinity X_1 is greater than the second electron affinity X_2 , and the third electron affinity X_3 is between the first and second electron affinities X_1 , X_2 . Hence, the electron affinity of the dielectric structure defined by the multiple dielectric layers **104** of FIG. **7** discretely decreases, and then discretely increases, from the bottom electrode **106** to the top electrode **108**. Further, the bottom conductive band edge of the dielectric structure steps up, and then steps down, from the bottom electrode **106** to the top electrode **108**. In alternative embodiments, the dielectric structure has other suitable electron affinity profiles and/or bottom conductive band edge profiles.

With reference to FIG. **9**, a cross-sectional view **900** of some alternative embodiments of the memory cell **102** of FIG. **6** is provided in which the memory cell **102** further comprises a cap layer **502**. The cap layer **502** may, for example, be as described with regard to FIG. **5**. Hence, the cap layer **502** may, for example, provide an enlarged switching window, reduced power consumption, other suitable benefits, or any combination of the foregoing.

With reference to FIG. **10**, a cross-sectional view **1000** of some embodiments of the memory cell **102** of FIG. **1** is provided in which the memory cell **102** is in an interconnect structure **1002** of an IC chip. The memory cell **102** underlies a top electrode wire **1004t** and a top electrode via **1006t**. Further, the memory cell **102** overlies a bottom electrode wire **1004b** and a bottom electrode via **1008**.

The top electrode via **1006t** extends downward from the top electrode wire **1004t** to the top electrode **108**. Further, the top electrode via **1006t** extends through a hard mask **1010** atop the top electrode **108**. In alternative embodiments, the hard mask **1010** is omitted. The hard mask **1010** may be or comprise, for example, silicon nitride and/or some other suitable dielectric(s). The top electrode wire **1004t** and the top electrode via **1006t** may be or comprise, for example, copper, aluminum, aluminum copper, some other suitable metal(s), or any combination of the foregoing.

The bottom electrode via **1008** extends upward from the bottom electrode wire **1004b** to the bottom electrode **106** and comprises a via plug **1008p** and a via liner **1008l**. In alternative embodiments, the via liner **1008l** is omitted. The bottom electrode wire **1004b** may be or comprise, for example, copper, aluminum, aluminum copper, some other suitable metal(s), or any combination of the foregoing.

The via liner **1008l** cups an underside of the via plug **1008p** to separate the via plug **1008p** from the bottom electrode wire **1004b**. In some embodiments, the via liner

1008l is an adhesion layer to enhance deposition of a layer from which the via plug **1008p** is formed. In some embodiments, the via liner **1008l** is a diffusion barrier to prevent material of the bottom electrode wire **1004b** from diffusing upward to the via plug **1008p** and/or the bottom electrode **106**. The via liner **1008l** may be or comprise, for example, tantalum nitride and/or some other suitable conductive barrier material(s). The via plug **1008p** may be or comprise, for example, titanium nitride, aluminum, titanium, tantalum, gold, platinum, tungsten, some other suitable conductive material(s), or any combination of the foregoing. In some embodiments, the via plug **1008p** is integrated with and/or is the same material as the bottom electrode **106**, such that there is no boundary between the bottom electrode **106** and the via plug **1008p**.

A dielectric structure surrounds the memory cell **102**, as well as the top electrode wire **1004t**, the top electrode via **1006t**, the bottom electrode wire **1004b**, and the bottom electrode via **1008**. The dielectric structure comprises the hard mask **1010** and further comprises a sidewall spacer structure **1012** on sidewall(s) of the top electrode **108**. The sidewall spacer structure **1012** may be or comprise, for example, silicon nitride and/or some other suitable dielectric(s). Additionally, the dielectric structure comprises a plurality of intermetal dielectric (IMD) layers **1014**, a via dielectric layer **1016**, an etch stop layer **1018**.

The IMD layers **1014** respectively surround the bottom electrode wire **1004b** and the top electrode wire **1004t**. The IMD layers **1014** may be or comprise, for example, an extreme low k dielectric and/or some other suitable dielectric(s).

The via dielectric layer **1016** and the etch stop layer **1018** are stacked between the IMD layers **1014**. The via dielectric layer **1016** surrounds the bottom electrode via **1008**, between the memory cell **102** and the bottom electrode wire **1004b**. The via dielectric layer **1016** may, for example, be or comprise silicon carbide, silicon-rich oxide, some other suitable dielectric(s), or any combination of the foregoing. The etch stop layer **1018** covers the via dielectric layer **1016** and wraps around a top of the memory cell **102**. The etch stop layer **1018** may be or comprise, for example, silicon carbide and/or some other suitable dielectric(s).

With reference to FIGS. **11A** and **11B**, cross-sectional views **1100A**, **1100B** of different alternative embodiments of the memory cell **102** of FIG. **10** are provided. In FIG. **11A**, the memory cell **102** further comprises a cap layer **502** as described with regard to FIG. **5**. Further, the sidewall spacer structure **1012** is on sidewalls of the cap layer **502**. In FIG. **11B**, the memory cell **102** is directly on the bottom electrode wire **1004b** and each individual layer of the memory cell **102**, except for the top electrode **108**, has a U-shaped or V-shaped profile. Other suitable profiles are, however, amenable in alternative embodiments. Further, relative positioning between features has been rearranged and the bottom electrode via **1008**, the sidewall spacer structure **1012**, and the hard mask **1010** are omitted.

The memory cell **102** of FIG. **11B** may be formed with a single photolithography/etching process. For example, the via dielectric layer **1016** and a bordering one of the IMD layers **1014** may be deposited over the bottom electrode wire **1004b**. The via dielectric layer **1016** and the IMD layer may then be patterned to define a memory cell opening exposing the bottom electrode wire **1004b**. Further, the individual layers from which the memory cell **102** is formed may be conformally deposited in the memory cell opening and covering the IMD layer. With the layers deposited, a planarization may be performed to uncover the IMD layer and

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to pattern the layers into the memory cell **102**. Because photolithography is expensive, forming the memory cell **102** of FIG. **11B** by a single photolithography/etching process may be a substantial cost savings.

With reference to FIG. **12A**, a cross-sectional view **1200A** of some embodiments of an IC chip comprising multiple memory cells **102** is provided in which the memory cells **102** are integrated into individual one-transistor one-resistor (1T1R) cells **1202** and are each configured as in FIG. **10**. The 1T1R cells **1202** comprise individual drain regions **1204** and individual drain-side conductive paths **1206**.

The drain regions **1204** are doped regions of a substrate **1208** and each has an opposite doping type as an adjoining region of the substrate **1208**. Further, the drain regions **1204** are electrically separated from each other by a trench isolation structure **1210** and partially define access transistors **1212** (partially shown) used to individually select the memory cells **102**. The trench isolation structure **1210** extends into a top of the substrate **1208** and comprises silicon oxide and/or some other suitable dielectric material (s). The trench isolation structure **1210** may, for example, be a shallow trench isolation (STI) structure or some other suitable trench isolation structure. The substrate **1208** may, for example, be a bulk silicon substrate, a silicon-on-insulator (SOI) substrate, or some other suitable semiconductor substrate.

The drain-side conductive paths **1206** electrically couple the drain regions **1204** to the memory cells **102** and are defined by the interconnect structure **1002**. The interconnect structure **1002** comprises a plurality of wires **1004** and a plurality of vias **1006**. The plurality of wires **1004** comprises the top electrode wires **1004t** and the bottom electrode wires **1004b**. In some embodiments, the top electrode wires **1004t** correspond to bit lines BL. The plurality of vias **1006** comprises the top electrode via **1006t**. A level of the vias **1006** nearest the substrate **1208** is in an interlayer dielectric (ILD) layer **1214**, whereas remaining levels of the vias **1006** and the wires **1004** are in IMD layers **1014**. The wires **1004** and the vias **1006** are conductive and may be or comprise, for example, copper, aluminum, aluminum copper, some other suitable conductive material(s), or any combination of the foregoing.

A peripheral region **1216** to a side of the 1T1R cells **1202** accommodates a peripheral device **1218** (only partially shown). The peripheral device **1218** comprises a pair of source/drain regions **1220** (only one of which is shown) in the substrate **1208**, and further comprises a gate structure (not shown) between the source/drain regions **1220**. The source/drain regions **1220** are doped regions of the substrate **1208** and each has an opposite doping type as an adjoining region of the substrate **1208**. The peripheral device **1218** may, for example, be a metal-oxide-semiconductor field-effect transistor (MOSFET) or some other suitable type of semiconductor device. In alternative embodiments, the peripheral device **1218** is a fin field-effect transistor (FinFET), a gate-all-around field-effect transistor (GAA FET), a nanowire field-effect transistor, a nanosheet field-effect transistor, or some other suitable type of semiconductor device.

With reference to FIG. **12B**, a cross-sectional view **1200B** of some embodiments of the IC chip of FIG. **12A** is provided along an axis orthogonal to an axis along which the cross-sectional view **1200A** of FIG. **12A** is taken. The 1T1R cells **1202** comprise individual memory cells **102**, individual drain-side conductive paths **1206**, individual access transistors **1212**, and individual source-side conductive paths **1222**.

The access transistors **1212** are on the substrate **1208**, between the substrate **1208** and the interconnect structure

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1002. Further, the access transistors **1212** are electrically separated from each other by the trench isolation structure **1210**. The access transistors **1212** comprise individual drain regions **1204**, individual source regions **1224**, individual gate dielectric layers **1226**, and individual gate electrodes **1228**. The gate electrodes **1228** respectively overlie the gate dielectric layers **1226** and, in some embodiments, define word lines WL. The drain and source regions **1204**, **1224** are doped regions of the substrate **1208** and each has an opposite doping type as an adjoining region (e.g., a bulk) of the substrate **1208**. The drain regions **1204** respectively border drain sides of the gate electrodes **1228**, and the source regions **1224** respectively border source sides of the gate electrodes **1228**. The access transistors **1212** may, for example, be MOSFETs or some other suitable type of semiconductor devices.

The drain-side conductive paths **1206** electrically couple the drain regions **1204** to the memory cells **102**, and the source-side conductive paths **1222** electrically couple the source regions **1224** to source lines SL. The drain-side and source-side conductive paths **1206**, **1222** are defined by the plurality of wires **1004** and the plurality of vias **1006**.

With reference to FIG. **13**, a top layout **1300** of some embodiments of the IC chip of FIGS. **12A** and **12B** is provided. The cross-sectional views **1200A**, **1200B** of FIGS. **12A** and **12B** may, for example, respectively be taken along lines C and D or other suitable locations. The IC chip comprises a plurality of memory cells **102** in a plurality of rows and a plurality of columns, thereby defining a memory array **1302**. The memory cells **102** may, for example, be as in any of FIG. **1**, **4A**, **4B**, **5-7**, **9**, **10**, **11A**, **11B**, **12A**, or **12B**.

Peripheral devices **1218** surround the memory array **1302** at a peripheral region **1216** of the IC chip. The peripheral devices **1218** may, for example, be or comprise transistors and/or other suitable semiconductor device(s). Further, the peripheral devices **1218** may, for example, implement ECC circuitry, read/write circuitry, other suitable circuitry for operating the memory cells **102**, or any combination of the foregoing.

As described above, the memory cells **102** include individual HEA dielectric layers arranged closest to individual bottom electrodes to reduce the likelihood of the memory cells **102** becoming stuck during cycling at least when the memory cells **102** are RRAM cells. Hence, the likelihood of hard reset/failure bits is reduced. Because the likelihood of hard reset/failure bits is reduced, ECC capacity in ECC circuitry for the memory array **1302** is less likely to be used by hard reset/failure bits. In some embodiments, the ECC circuitry is wholly or partially defined by a portion **1304** of the peripheral region **1216**. Because ECC capacity is less likely to be used by hard reset/failure bits, the likelihood of the memory array **1302** failing is less likely. Because the likelihood of the memory array **1302** failing is reduced by the HEA dielectric layers, no additional ECC capacity and therefore no additional IC chip area are needed.

With reference to FIG. **14**, a schematic diagram **1400** of some embodiments of a 1T1R cell **1202** in FIGS. **12A** and **12B** is provided. The 1T1R cell **1202** is representative of each other 1T1R cell **1202** in FIGS. **12A** and **12B** and includes a memory cell **102** and an access transistor **1212** that are electrically coupled in series from a bit line BL to a source line SL. As described above, the memory cell **102** includes a HEA dielectric layer **104h** to prevent the memory cell **102** from becoming stuck during cycling at least when the memory cell **102** is RRAM. The access transistor **1212** is gated by a word line WL and selectively couples the memory cell **102** to the source line SL depending on a signal

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carried on the word line WL. The access transistor **1212** is illustrated as a MOSFET but may be some other suitable type of semiconductor device.

With reference to FIGS. **15A-15C**, schematic diagrams **1500A-1500C** of some different alternative embodiments of the 1T1R cell **1202** of FIG. **14** is provided in which the access transistor **1212** is a different type of semiconductor device. In FIG. **15A**, the access transistor **1212** is a bipolar junction transistor (BJT). In FIG. **15B**, the access transistor **1212** is a HEMT with a Schottky gate contact. In FIG. **15C**, the access transistor **1212** is a metal-oxide-semiconductor (MOS) high-electron-mobility transistor (HEMT). In alternative embodiments, the access transistor **1212** is some other suitable type of semiconductor device.

While the memory cells **102** of FIGS. **10**, **11A**, **11B**, **12A**, **12B**, **14**, and **15A-15C** are illustrated with a single LEA dielectric layer **104**, the memory cells **102** of FIGS. **10**, **11A**, **11B**, **12A**, **12B**, **14**, and **15A-15C** may have two or more LEA dielectric in alternative embodiments. Examples are illustrated in FIGS. **5** and **6**. While the memory cells **102** of FIGS. **11B**, **12A**, **12B**, **14**, and **15A-15C** are illustrated without the cap layer **502** of FIG. **5**, the memory cells **102** of FIGS. **11B**, **12A**, **12B**, **14**, and **15A-15C** may have the cap layer **502** in alternative embodiments. While each memory cell **102** in the IC chip of FIGS. **12A** and **12B** is configured according to the embodiments of FIG. **10**, the embodiments of FIGS. **11A** and **11B** may alternatively be employed. While the 1T1R cells **1202** of FIGS. **14** and **15A-15C** comprise the memory cell **102** of FIG. **1**, the 1T1R cells **1202** of FIGS. **14** and **15A-15C** may instead comprise the memory cell **102** in any of FIG. **4A**, **4B**, **5-7**, **9**, **10**, **11A**, **11B**, **12A**, or **12B** in alternative embodiments.

With reference to FIGS. **16-23**, a series of cross-sectional views **1600-2300** of some embodiments of a method for forming memory cells integrated with 1T1R cells and comprising individual HEA dielectric layers at corresponding bottom electrodes is provided. The cross-sectional views **1600-2300** may, for example, be taken along line C in FIG. **13** or at some other suitable location in FIG. **13**. Further, the cross-sectional views **1600-2300** may, for example, illustrate formation of the IC chip of FIGS. **12A** and **12B** with the addition of a cap layer.

As illustrated by the cross-sectional view **1600** of FIG. **16**, a trench isolation structure **1210** is formed extending into a top of a substrate **1208**. The trench isolation structure **1210** individually surrounds and demarcates regions of the substrate **1208** at which 1T1R cells **1202** are being formed. Further, the trench isolation structure **1210** surrounds and demarcates a peripheral region **1216** of the IC chip.

Also illustrated by the cross-sectional view **1600** of FIG. **16**, a plurality of semiconductor devices is formed on the substrate **1208**. The plurality of semiconductor devices comprises access transistors **1212** (only partially shown) individual to and respectively at the 1T1R cells **1202** being formed. Further, the plurality of semiconductor devices comprises a peripheral device **1218** (only partially shown) at the peripheral region **1216** of the IC chip. The access transistors **1212** comprise individual drain regions **1204** and individual source regions (not shown) in the substrate **1208**. Further, the access transistors **1212** comprise individual gate structures (not shown). The gate structures have individual drain sides respectively bordering the drain regions **1204** and further have individual source sides respectively bordering the source regions. The peripheral device **1218** comprises a pair of source/drain regions **1220** (only one of which

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is shown) in the substrate **1208** and further comprises a gate structure (not shown) between and bordering the source/drain regions **1220**.

As illustrated by the cross-sectional view **1700** of FIG. **17**, an interconnect structure **1002** is partially formed over and electrically coupled to the semiconductor devices (e.g., the access transistors **1212** and the peripheral device **1218**). The interconnect structure **1002** comprises a plurality of wires **1004** and a plurality of vias **1006** stacked in a dielectric structure. The dielectric structure comprises an ILD layer **1214** and a plurality of IMD layers **1014** over the ILD layer **1214**. The plurality of wires **1004** comprises a plurality of bottom electrode wires **1004b** along a top surface of the interconnect structure **1002**. The bottom electrode wires **1004b** are individual to and respectively at the 1T1R cells **1202** being formed. Further, the bottom electrode wires **1004b** are respectively electrically coupled to the drain regions **1204** of the access transistors **1212** by the wires **1004** and the vias **1006**.

Also illustrated by the cross-sectional view **1700** of FIG. **17**, a via dielectric layer **1016** is deposited on the interconnect structure **1002**.

As illustrated by the cross-sectional view **1800** of FIG. **18**, bottom electrode vias **1008** are formed in the via dielectric layer **1016**. Note that for drawing compactness, a lower portion of the interconnect structure **1002** and structure underlying the lower portion are omitted herein and in subsequent figures. The bottom electrode vias **1008** are individual to the 1T1R cells **1202** being formed and extend through the via dielectric layer **1016** respectively to the bottom electrode wires **1004b**. Further, the bottom electrode vias **1008** comprise individual via plugs **1008p** and individual via liners **1008l**. In alternative embodiments, the via liners **1008l** are omitted. The via liners **1008l** cup undersides of the via plugs **1008p** to separate the via plugs **1008p** from the bottom electrode wires **1004b**.

A process for forming the bottom electrode vias **1008** may, for example, comprise: 1) patterning the via dielectric layer **1016** to form openings individual to and respectively exposing the bottom electrode wires **1004b**; 2) depositing a liner layer and a plug layer filling the openings and covering the via dielectric layer **1016**; and 3) performing a planarization into the plug and liner layers to uncover the via dielectric layer **1016**. Other suitable processes are, however, amenable.

Also illustrated by the cross-sectional view **1800** of FIG. **18**, a bottom electrode layer **1802** is deposited covering the via dielectric layer **1016** and the bottom electrode vias **1008**. The bottom electrode layer **1802** is conductive and is electrically coupled to the bottom electrode wires **1004b** through the bottom electrode vias **1008**. The bottom electrode layer **1802** may, for example, be or comprise a metal, a metal nitride, a metal oxide, doped polysilicon, some other suitable conductive material(s), or any combination of the foregoing. Further, the bottom electrode layer **1802** may, for example, be or comprise aluminum, titanium, tantalum, gold, platinum, tungsten, nickel, iridium, titanium nitride, tantalum nitride, N-doped polysilicon, P-doped polysilicon, some other suitable material(s), or any combination of the foregoing. In some embodiments, the bottom electrode layer **1802** is the same material as the via plugs **1008p**.

While the bottom electrode layer **1802** and the bottom electrode vias **1008** are described as being separately formed, the bottom electrode layer **1802** and the bottom electrode vias **1008** may be formed together in alternative embodiments. In at least some of these alternative embodiments, the via plugs **1008p** are part of the bottom electrode

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layer **1802** and there is no boundary between the via plugs **1008p** and the bottom electrode layer **1802**.

As illustrated by the cross-sectional view **1900** of FIG. **19**, multiple dielectric layers **104** are deposited vertically stacked over the bottom electrode layer **1802**. Further, a dielectric layer **104h** having a highest electron affinity amongst the multiple dielectric layers **104** (i.e., a HEA dielectric layer **104h**) is deposited first and is hence closest to the bottom electrode layer **1802**. As will be seen hereafter, the multiple dielectric layers **104** are divided into segments that are individual to the memory cells being formed and that serve as switching layers for the memory cells. Further, it has been appreciated that by arranging the HEA dielectric layer **104h** closest to the bottom electrode layer **1802**, the likelihood of the memory cells becoming stuck during cycling is reduced at least when the memory cells are RRAM cells.

The multiple dielectric layers **104** include the HEA dielectric layer **104h** and further include a dielectric layer **104l** having a low electron affinity (i.e., a LEA dielectric layer **104l**) compared to the HEA dielectric layer **104h**. The LEA dielectric layer **104l** overlies the HEA dielectric layer **104h** and is hence farther from the bottom electrode layer **1802** than the HEA dielectric layer **104h**. In alternative embodiments, the multiple dielectric layers **104** include two or more LEA dielectric layers vertically stacked over the HEA dielectric layer **104h** and each as the LEA dielectric layer **104l** is described.

The HEA dielectric layer **104h** is a different dielectric material than the LEA dielectric layer **104l**. Each of the HEA and LEA dielectric layers **104h**, **104l** may, for example, be a metal oxide, a metal oxynitride, a component metal oxide, some other suitable dielectric(s), or any combination of the foregoing. Further, each of the HEA and LEA dielectric layers **104h**, **104l** may, for example, be titanium oxide (e.g., TiO_2), hafnium oxide (e.g., HfO_2), hafnium aluminum oxide (e.g., $\text{Hf}_x\text{Al}_{1-x}\text{O}_2$), tantalum oxide (e.g., Ta_2O_5), hafnium tantalum oxide (e.g., $\text{Hf}_x\text{Ta}_{1-x}\text{O}_2$), tungsten oxide (e.g., WO_2), zirconium oxide (e.g., ZrO_2), aluminum oxide (e.g., Al_2O_3), sulfated tin oxide (e.g., STO), some other suitable dielectric(s), or any combination of the foregoing. In some embodiments, the HEA dielectric layer **104h** is a high k dielectric and/or the LEA dielectric layer **104l** is a high k dielectric.

The HEA and LEA dielectric layers **104h**, **104l** have different material systems or different material compositions. Different material systems correspond to different sets of elements. For example, the HEA dielectric layer **104h** may be tantalum oxide (e.g., Ta_2O_5), whereas the LEA dielectric layer **104l** may be aluminum oxide (e.g., Al_2O_3). Different material compositions correspond to different ratios of elements for the same set of elements (e.g., the same material systems). For example, the HEA dielectric layer **104h** may be aluminum oxide (e.g., Al_2O_3) and the LEA dielectric layer **104l** may be aluminum oxide with a different ratio of aluminum and oxide (e.g., Al_xO_y , where $x \neq 2$ and $y \neq 3$). While the preceding two examples provide specific materials, other suitable materials are, however, amenable.

The HEA and LEA dielectric layers **104h**, **104l** have individual thicknesses T_d . In some embodiments, the thicknesses T_d are about 1-50 nanometers, about 1-25 nanometers, about 25-50 nanometers, or some other suitable value. If the thickness T_d of the HEA or LEA dielectric layer **104h**, **104l** is too small (e.g., less than about 1 nanometer or some other suitable value), benefits from material properties of the dielectric layer may not be attained. For example, if the

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thickness T_d of the HEA dielectric layer **104h** is too small, the HEA dielectric layer **104h** may not prevent memory cells being formed from becoming stuck during cycling at least when the memory cells are RRAM cells. If the thickness T_d of the HEA or LEA dielectric layer **104h**, **104l** is too large (e.g., more than about 50 nanometers or some other suitable value), operating voltages of the memory cells being formed may be too high. The high voltages may, for example, increase power consumption, reduce the lifespan of the memory cell, and increase the risk of device failure.

A process for depositing the multiple dielectric layers **104** may, for example, comprise sequentially depositing the multiple dielectric layers **104** by chemical vapor deposition (CVD), physical vapor deposition (PVD), some other suitable deposition process(es), or any combination of the foregoing. In some embodiments, the HEA dielectric layer **104h** is deposited by thermal oxidation of the bottom electrode layer **1802** and then the LEA dielectric layer **104l** is deposited by CVD, PVD, or some other suitable deposition process.

As noted above, the HEA and LEA dielectric layers **104h**, **104l** have different material systems or different material compositions. In some embodiments in which the HEA and LEA dielectric layers **104h**, **104l** have different material systems, the HEA and LEA dielectric layers **104h**, **104l** are deposited by vapor deposition respectively using different sets of precursors. In some embodiments in which the HEA and LEA dielectric layers **104h**, **104l** have different material compositions, the HEA and LEA dielectric layers **104h**, **104l** are deposited by vapor deposition using the same set of precursors but different ratios of the precursors. Further, in some embodiments in which the HEA and LEA dielectric layers **104h**, **104l** have different material compositions, the HEA and LEA dielectric layers **104h**, **104l** are deposited in situ within a common process chamber.

Also illustrated by the cross-sectional view **1900** of FIG. **19**, a cap layer **502** and a top electrode layer **1902** are deposited vertically stacked over the multiple dielectric layers **104**. In alternative embodiments, the cap layer **502** is omitted and is hence not deposited.

The cap layer **502** has a high affinity for oxygen compared to the bottom and top electrode layers **1802**, **1902**. In other words, the cap layer **502** depends upon less energy to react with oxygen than the bottom and top electrode layers **1802**, **1902**. At least when the memory cells being formed are metal-ion type RRAM cells and oxygen-ion type RRAM cells, the cap layer **502** may enhance performance of the memory cells. For example, the cap layer **502** may increase the switching windows and/or reduce operating voltages. The cap layer **502** may, for example, be or comprise aluminum, titanium, tantalum, hafnium, titanium oxide, hafnium oxide, zirconium oxide, germanium oxide, cerium oxide, some other suitable material(s), or any combination of the foregoing. In some embodiments, the cap layer **502** is conductive and/or is metal. For example, in embodiments in which the memory cells being formed are a metal-ion type RRAM cells, the cap layer **502** is metal. In alternative embodiments, the cap layer **502** is dielectric. In embodiments in which the cap layer **502** is dielectric, the cap layer **502** has a lower electron affinity than the HEA dielectric layer **104h**.

The top electrode layer **1902** may, for example, be or comprise a metal, a metal nitride, doped polysilicon, some other suitable conductive material(s), or any combination of the foregoing. Further, the top electrode layer **1902** may, for example, be or comprise aluminum, titanium, tantalum, gold, platinum, tungsten, nickel, iridium, titanium nitride,

tantalum nitride, N-doped polysilicon, P-doped polysilicon, some other suitable material(s), or any combination of the foregoing.

As illustrated by the cross-sectional view **2000** of FIG. **20**, hard masks **1010** are formed individual to and respectively at the 1T1R cells **1202** being formed. As seen hereafter, the hard masks **1010** have patterns for memory cells of the 1T1R cells **1202**. The hard masks **1010** may, for example, be formed by depositing a hard mask layer over the top electrode layer **1902** (see, e.g., FIG. **19**) and subsequently patterning the hard mask layer into the hard masks **1010**. The patterning may, for example, be performed by a photolithography/etching process or some other suitable patterning process.

Also illustrated by the cross-sectional view **2000** of FIG. **20**, a first etch is performed into the top electrode layer **1902** (see, e.g., FIG. **19**) and the cap layer **502** (see, e.g., FIG. **19**) with the hard masks **1010** in place. The first etch stops on the multiple dielectric layers **104** and transfers patterns of the hard masks **1010** to the top electrode layer **1902** and the cap layer **502**. By transferring the patterns to the top electrode layer **1902**, the first etch divides the top electrode layer **1902** into top electrodes **108** individual to the memory cells being formed. By transferring the patterns to the cap layer **502**, the first etch divides the cap layer **502** into cap segments individual to the memory cells.

As illustrated by the cross-sectional view **2100** of FIG. **21**, sidewall spacer structures **1012** are formed on common sidewalls defined by the hard masks **1010**, the top electrodes **108**, and the cap layer **502**. The sidewall spacer structures **1012** may, for example, be or comprise silicon nitride and/or some other suitable dielectric(s). A process for forming the sidewall spacer structures **1012** may, for example, comprise: 1) a depositing a spacer layer covering and conformably lining the structure of FIGS. **20**; and 2) etching back the spacer layer. Other suitable processes are, however, amenable.

Also illustrated by the cross-sectional view **2100** of FIG. **21**, a second etch is performed into the multiple dielectric layers **104** and the bottom electrode layer **1802** with the hard masks **1010** and the sidewall spacer structures **1012** in place. The second etch stops on the via dielectric layer **1016** and transfers patterns collectively defined by the hard masks **1010** and the sidewall spacer structures **1012** to the dielectric layers **104** and the bottom electrode layer **1802**. By transferring the patterns to the multiple dielectric layers **104**, the second etch divides the multiple dielectric layers **104** into dielectric segments individual to the memory cells being formed. By transferring the patterns to the bottom electrode layer **1802**, the second etch divides the bottom electrode layer **1802** into bottom electrodes **106** individual to the memory cells.

Upon completion of the second etch, memory cells **102** individual to the 1T1R cells **1202** being formed remain. Individual bottom electrodes **106** of the memory cells **102** respectively overlie and electrically couple to the bottom electrode vias **1008**. Individual dielectric segments of the multiple dielectric layers **104** respectively overlie the bottom electrodes **106**. Individual cap segments of the cap layer **502** respectively overlie the dielectric segments. Individual top electrodes **108** respectively overlying the cap segments. The memory cells **102** may, for example, be oxygen-ion type RRAM cells, metal-ion type RRAM cells, or some other suitable type of memory cells.

During operation of the memory cells **102**, the individual dielectric segments serve as switching layers that change between HRSs and LRSs. Further, because the HEA dielec-

tric layer **104h** is closest to the bottom electrodes **106** amongst the multiple dielectric layers **104**, the likelihood of the memory cells **102** becoming stuck during cycling is reduced at least when the memory cells are RRAM cells. Hence, the likelihood of hard reset/failure bits is reduced. Because the likelihood of hard reset/failure bits is reduced, ECC capacity for a memory array (not shown) incorporating the memory cells **102** is less likely to be used by hard reset/failure bits. Hence, the likelihood of the memory array failing is less likely. Because the likelihood of the memory array failing is reduced by the HEA dielectric layer **104h**, no additional ECC capacity and therefore no additional IC chip area are needed.

Because the likelihood of the memory cells **102** becoming stuck during cycling is reduced by the HEA dielectric layer **104h**, the likelihood may be reduced by an additional deposition process during the method. As such, the HEA dielectric layer **104h** adds little to no extra cost and is compatible with 40 nanometer process nodes and smaller.

As illustrated by the cross-sectional view **2200** of FIG. **22**, the interconnect structure **1002** is completed around the memory cells **102**. This includes depositing an etch stop layer **1018** and an additional IMD layer **1014** covering the memory cells **102**, and subsequently forming a plurality of additional vias **1004** and a plurality of additional vias **1006** in the etch stop layer **1018** and the additional IMD layer **1014**. The plurality of additional vias **1004** comprises top electrode vias **1004t** respectively overlying the memory cells **102**, and the plurality of additional vias **1006** comprises top electrode vias **1006t** extending from the top electrode vias **1004t** respectively to the top electrodes **108** of the memory cells **102**.

As illustrated by the cross-sectional view **2300** of FIG. **23**, conductive filaments **402** individual to the memory cells **102** are formed in the multiple dielectric layers **104** respectively at the memory cells **102**. The forming may, for example, comprise applying a forming voltage across each of the memory cells **102**. For example, the bottom electrodes **106** of the memory cells **102** may be grounded while the top electrodes **108** of the memory cells **102** are biased. Other processes for forming the conductive filaments **402** are, however, amenable.

While FIGS. **16-23** are described with reference to a method, it will be appreciated that the structures shown in FIGS. **16-23** are not limited to the method but rather may stand alone separate of the method. While FIGS. **16-23** are described as a series of acts, it will be appreciated that the order of the acts may be altered in other embodiments. While FIGS. **16-23** illustrate and describe as a specific set of acts, some acts that are illustrated and/or described may be omitted in other embodiments. Further, acts that are not illustrated and/or described may be included in other embodiments.

With reference to FIG. **24**, a block diagram **2400** of some embodiments of the method of FIGS. **16-23** is provided.

At **2402**, an access transistor is formed on a substrate. See, for example, FIG. **16**.

At **2404**, an interconnect structure is partially formed over the substrate and the access transistor, where the interconnect structure comprises a bottom electrode wire overlying and electrically coupled to the access transistor. See, for example, FIG. **17**.

At **2406**, a via dielectric layer is formed over the interconnect structure. See, for example, FIG. **17**.

At **2408**, a bottom electrode via is formed extending through the via dielectric layer to the bottom electrode wire. See, for example, FIG. **18**.

At **2410**, a bottom electrode layer is deposited over the via dielectric layer and the bottom electrode via. See, for example, FIG. **18**.

At **2412**, multiple dielectric layers are deposited vertically stacked over the bottom electrode layer, where a dielectric layer having a highest electron affinity amongst the multiple dielectric layers is deposited first and is closest to the bottom electrode layer. See, for example, FIG. **19**.

At **2414**, a cap layer is deposited over the multiple dielectric layers. See, for example, FIG. **19**.

At **2416**, a top electrode layer is deposited over the cap layer. See, for example, FIG. **19**.

At **2418**, the top and bottom electrode layers, the cap layer, and the multiple dielectric layers are patterned to form a memory cell overlying and electrically coupled to the bottom electrode wire through the bottom electrode via. See, for example, FIGS. **20** and **21**.

At **2420**, the interconnect structure is completed around the memory cell. See, for example, FIG. **22**.

At **2422**, a forming voltage is applied across the memory cell to form a conductive filament in the multiple dielectric layers. See, for example, FIG. **23**.

While the block diagram **2400** of FIG. **24** is illustrated and described herein as a series of acts or events, it will be appreciated that the illustrated ordering of such acts or events is not to be interpreted in a limiting sense. For example, some acts may occur in different orders and/or concurrently with other acts or events apart from those illustrated and/or described herein. Further, not all illustrated acts may be required to implement one or more aspects or embodiments of the description herein, and one or more of the acts depicted herein may be carried out in one or more separate acts and/or phases.

With reference to FIGS. **25-29** a series of cross-sectional views **2500-2900** of some alternative embodiments of the method of FIGS. **16-23** is provided in which layers from which the memory cells are formed are patterned using a planarization. The cross-sectional views **2500-2900** may, for example, be taken along line C in FIG. **13** or at some other suitable location in FIG. **13**. Further, the cross-sectional views **2500-2900** may, for example, illustrate formation of the memory cell **102** of FIG. **11B** with the addition of a cap layer.

As illustrated by the cross-sectional view **2500** of FIG. **25**, the acts at FIGS. **16** and **17** are performed. A trench isolation structure **1210** is formed extending into a top of a substrate **1208** as illustrated and described with regard to FIG. **16**. Further, a plurality of semiconductor devices is formed on the substrate **1208** as illustrated and described with regard to FIG. **16**. The plurality of semiconductor devices comprises access transistors **1212** (only partially shown) and a peripheral device **1218** (only partially shown). An interconnect structure **1002** is partially formed over and electrically coupled to the semiconductor devices as illustrated and described with regard to FIG. **17**. Further, a via dielectric layer **1016** is deposited on the interconnect structure **1002** as illustrated and described with regard to FIG. **17**.

Also illustrated by the cross-sectional view **2500** of FIG. **25**, an additional IMD layer **1014** is deposited covering the via dielectric layer **1016**.

As illustrated by the cross-sectional view **2600** of FIG. **26**, the via dielectric layer **1016** and the additional IMD layer **1014** atop the via dielectric layer **1016** are patterned to form memory cell openings **2602** individual to and respectively at the 1T1R cells **1202** being formed. Note that for drawing compactness, a lower portion of the interconnect structure **1002** and structure underlying the lower portion are omitted

herein and in subsequent figures. The patterning may, for example, be performed by a photolithography/etching process or some other suitable patterning process.

As illustrated by the cross-sectional view **2700** of FIG. **27**, a bottom electrode layer **1802**, multiple dielectric layers **104**, a cap layer **502**, and a top electrode layer **1902** are deposited lining and filling the memory cell openings **2602** (see, e.g., FIG. **26**). In alternative embodiments, the cap layer **502** is omitted and is hence not formed. In alternative embodiments, the multiple dielectric layers **104** include three or more dielectric layers. The bottom electrode layer **1802**, the multiple dielectric layers **104**, the cap layer **502**, and the top electrode layer **1902** are as described with regard to FIGS. **18** and **19**. Hence, a dielectric layer **104h** having a highest electron affinity amongst the multiple dielectric layers **104** (i.e., a HEA dielectric layer **104h**) is deposited first and is hence closest to the bottom electrode layer **1802**. It has been appreciated that by arranging the HEA dielectric layer **104h** closest to the bottom electrode layer **1802**, the likelihood of memory cells becoming stuck during cycling is reduced. The bottom electrode layer **1802**, the multiple dielectric layers **104**, the cap layer **502**, and the top electrode layer **1902** are formed as described with regard to FIGS. **18** and **19**.

As illustrated by the cross-sectional view **2800** of FIG. **28**, a planarization is performed into the bottom electrode layer **1802** (see, e.g., FIG. **27**), the multiple dielectric layers **104**, the cap layer **502**, and the top electrode layer **1902** (see, e.g., FIG. **27**). The planarization stops on the additional IMD layer **1014** atop the via dielectric layer **1016** and transfers a pattern of the memory cell openings **2602** (see, e.g., FIG. **26**) to the bottom electrode layer **1802**, the multiple dielectric layers **104**, the cap layer **502**, and the top electrode layer **1902**.

By transferring the pattern, the planarization form memory cells **102** individual to and respectively at the 1T1R cells **1202**. Transferring the pattern to the bottom and top electrode layers **1802**, **1902** divides the bottom and top electrode layers **1802**, **1902** respectively into bottom electrodes **106** individual to the memory cells **102** and top electrodes **108** individual to the memory cells **102**. Transferring the pattern to the multiple dielectric layers **104** divides the multiple dielectric layers **104** into dielectric segments individual to the memory cells **102**. Transferring the pattern to the cap layer **502** divides the cap layer **502** into cap segments individual to the memory cells **102**. Upon completion of the patterning, the bottom electrodes **106**, the dielectric segments, and the cap segments have U-shaped profiles. V-shaped and other suitable profiles are, however, amenable.

Because the planarization transfers a pattern from the patterning at FIG. **26** to the various layers of the memory cells **102**, the memory cells **102** may be formed by a single photolithography/etching process. This is in contrast to the multiple photolithography/etching processes (e.g., respectively at FIGS. **18** and **20**) that may be used to form the memory cells **102** at FIGS. **16-23**. Because photolithography is expensive, reducing the number of photolithography/etching processes may significantly reduce costs.

As illustrated by the cross-sectional view **2900** of FIG. **29**, the acts at FIGS. **22** and **23** are performed. The interconnect structure **1002** is completed around the memory cells **102** as illustrated and described with regard to FIG. **22**. Conductive filaments **402** individual to the memory cells **102** are formed in the multiple dielectric layers **104** respectively at the memory cells **102** as illustrated and described with regard to FIG. **23**.

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While FIGS. 25-29 are described with reference to a method, it will be appreciated that the structures shown in FIGS. 25-29 are not limited to the method but rather may stand alone separate of the method. While FIGS. 25-29 are described as a series of acts, it will be appreciated that the order of the acts may be altered in other embodiments. While FIGS. 25-29 illustrate and describe as a specific set of acts, some acts that are illustrated and/or described may be omitted in other embodiments. Further, acts that are not illustrated and/or described may be included in other embodiments.

With reference to FIG. 30, a block diagram 3000 of some embodiments of the method of FIGS. 25-29 is provided.

At 3002, an access transistor is formed on a substrate. See, for example, FIGS. 16 and 25.

At 3004, an interconnect structure is partially formed over the substrate and the access transistor, where the interconnect structure comprises a bottom electrode wire overlying and electrically coupled to the access transistor. See, for example, FIGS. 17 and 25.

At 3006, a via dielectric layer is formed over the interconnect structure. See, for example, FIGS. 17 and 25.

At 3008, an IMD layer is formed over the via dielectric layer. See, for example, FIG. 25.

At 3010, the via dielectric layer and the IMD layer are patterned to form a memory cell opening overlying and exposing the bottom electrode wire. See, for example, FIG. 26.

At 3012, a bottom electrode layer is deposited covering the IMD layer and lining the memory cell opening. See, for example, FIG. 27.

At 3014, multiple dielectric layers are deposited vertically stacked, covering the bottom electrode layer, and lining the memory cell opening, where a dielectric layer having a highest electron affinity amongst the multiple dielectric layers is deposited first and is closest to the bottom electrode layer. See, for example, FIG. 27.

At 3016, a cap layer is deposited covering the multiple dielectric layers and lining the memory cell opening. See, for example, FIG. 27.

At 3018, a top electrode layer is deposited covering the cap layer and lining the memory cell opening. See, for example, FIG. 27.

At 3020, a planarization is performed into the top and bottom electrode layers, the cap layer, and the multiple dielectric layers to form a memory cell overlying and electrically coupled to the wire in the memory cell opening. See, for example, FIG. 28.

At 3022, the interconnect structure is completed around the memory cell. See, for example, FIG. 29.

At 3024, a forming voltage is applied across the memory cell to form a conductive filament in the multiple dielectric layers. See, for example, FIG. 29.

While the block diagram 3000 of FIG. 30 is illustrated and described herein as a series of acts or events, it will be appreciated that the illustrated ordering of such acts or events is not to be interpreted in a limiting sense. For example, some acts may occur in different orders and/or concurrently with other acts or events apart from those illustrated and/or described herein. Further, not all illustrated acts may be required to implement one or more aspects or embodiments of the description herein, and one or more of the acts depicted herein may be carried out in one or more separate acts and/or phases.

In some embodiments, the present disclosure provides a memory cell including: a bottom electrode; a top electrode overlying the bottom electrode; and a dielectric stack includ-

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ing a plurality of dielectric layers stacked between the bottom and top electrodes; wherein the plurality of dielectric layers includes a first dielectric layer, and the first dielectric layer is a closest one of the dielectric layers to the bottom electrode and has a highest electron affinity amongst the dielectric layers. In some embodiments, the plurality of dielectric layers includes a second dielectric layer overlying the first dielectric layer, wherein the second dielectric layer has a different set of elements than the first dielectric layer. In some embodiments, the plurality of dielectric layers includes a second dielectric layer overlying the first dielectric layer, wherein the second dielectric layer has a same set of elements as the first dielectric layer and further has a different ratio of the elements as the first dielectric layer. In some embodiments, the dielectric stack consists of two dielectric layers. In some embodiments, the memory cell further includes a cap layer overlying the dielectric stack, between the dielectric stack and the top electrode, wherein the cap layer has a higher affinity for oxygen than the top and bottom electrodes. In some embodiments, the plurality of dielectric layers includes a second dielectric layer and a third dielectric layer, wherein the second dielectric layer is between the first and third dielectric layers and has an electron affinity between that of the first dielectric layer and that of the third dielectric layer. In some embodiments, the plurality of dielectric layers includes a second dielectric layer and a third dielectric layer, wherein the second dielectric layer is between the first and third dielectric layers, and wherein the third dielectric layer has an electron affinity between that of the first dielectric layer and that of the second dielectric layer. In some embodiments, the memory cell further includes a conductive filament in the dielectric stack, wherein the conductive filament includes oxygen vacancies. In some embodiments, the memory cell further includes a conductive filament in the dielectric stack, wherein the conductive filament includes metal.

In some embodiments, the present disclosure provides a memory device including a memory cell, wherein the memory cell includes: a bottom electrode; a dielectric structure overlying the bottom electrode and including multiple different dielectric materials from top to bottom; and a top electrode overlying the dielectric structure; wherein the multiple different dielectric materials include a first dielectric material at the bottom electrode, and the first dielectric material has a bottom conductive band edge that is lowest amongst the multiple different dielectric materials. In some embodiments, a bottom conductive band edge of the dielectric structure steps up from the bottom electrode to a top surface of the dielectric structure. In some embodiments, a bottom conductive band edge of the dielectric structure steps up from the bottom electrode to a midpoint between the bottom electrode and a top surface of the dielectric structure, and wherein the bottom conductive band edge steps down from the midpoint to the top surface. In some embodiments, the memory cell further includes: a cap layer between and directly contacting the top electrode and the dielectric structure, wherein the cap layer depends upon less energy to react with oxygen than the top electrode. In some embodiments, the memory device further includes a conductive filament in the dielectric structure, wherein the conductive filament extends from a top surface of the dielectric structure towards a bottom surface of the dielectric structure and terminates before the bottom surface.

In some embodiments, the present disclosure provides a method including: depositing a bottom electrode layer over a substrate; depositing a dielectric film over and directly on the bottom electrode layer, wherein the dielectric film

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includes multiple different dielectric layers that are vertically stacked, wherein the multiple different dielectric layers include a first dielectric layer at the bottom electrode layer, and wherein the first dielectric layer has a highest electron affinity amongst the multiple different dielectric layers; depositing a top electrode layer over the dielectric film; and patterning the bottom electrode layer, the dielectric film, and the top electrode layer into a memory cell. In some embodiments, the method further includes depositing a cap layer over the dielectric film, wherein the top electrode layer is deposited over the cap layer and has a lesser affinity for oxygen than the cap layer. In some embodiments, the dielectric film includes a second dielectric layer, and wherein the depositing of the dielectric film includes: depositing the first dielectric layer by vapor deposition; and depositing the second dielectric layer over the first dielectric layer by vapor deposition, wherein the depositing of the first dielectric layer and the depositing of the second dielectric layer are performed using a same set of precursors but different ratios of the precursors. In some embodiments, the dielectric film includes a second dielectric layer, and wherein the depositing of the dielectric film includes: depositing the first dielectric layer by vapor deposition with a first set of precursors; and depositing the second dielectric layer over the first dielectric layer by vapor deposition with a second set of precursors different than the first set of precursors. In some embodiments, the method further includes applying a forming voltage having a positive polarity from a top electrode of the memory cell to a bottom electrode of the memory cell to form a conductive filament in a dielectric structure separating the top and bottom electrodes. In some embodiments, an electron affinity of the dielectric film decreases discretely and uninterrupted from a bottom surface of the dielectric film to a top surface of the dielectric film.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A memory cell, comprising:

a bottom electrode;

a top electrode overlying the bottom electrode; and

a dielectric stack comprising a plurality of dielectric layers stacked between the bottom and top electrodes, wherein the plurality of dielectric layers comprises a first dielectric layer, a second dielectric layer, and a third dielectric layer,

the second dielectric layer is between the first and third dielectric layers, and

the first dielectric layer is a closest one of the plurality of dielectric layers to the bottom electrode and has a highest electron affinity amongst the plurality of dielectric layers.

2. The memory cell according to claim 1, wherein the second dielectric layer has a different set of elements than the first dielectric layer.

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3. The memory cell according to claim 1, wherein the second dielectric layer has a same set of elements as the first dielectric layer and further has a different ratio of the elements than the first dielectric layer.

4. The memory cell according to claim 1, wherein the dielectric stack consists of three dielectric layers.

5. The memory cell according to claim 1, further comprising:

a cap layer overlying the dielectric stack, between the dielectric stack and the top electrode, wherein the cap layer has a higher affinity for oxygen than the top and bottom electrodes.

6. The memory cell according to claim 1, wherein the second dielectric layer has an electron affinity between that of the first dielectric layer and that of the third dielectric layer.

7. The memory cell according to claim 1, wherein the third dielectric layer has an electron affinity between that of the first dielectric layer and that of the second dielectric layer.

8. The memory cell according to claim 1, further comprising:

a conductive filament in the dielectric stack, wherein the conductive filament comprises oxygen vacancies.

9. The memory cell according to claim 1, further comprising:

a conductive filament in the dielectric stack, wherein the conductive filament comprises metal.

10. A memory device comprising a memory cell, wherein the memory cell comprises:

a bottom electrode;

a dielectric structure overlying the bottom electrode and comprising multiple different dielectric materials from top to bottom; and

a top electrode overlying the dielectric structure,

wherein the multiple different dielectric materials comprise at least three different dielectric materials, which comprise a first dielectric material at the bottom electrode,

the first dielectric material has a bottom conductive band edge that is lowest amongst the multiple different dielectric materials, and

a bottom conductive band edge of the dielectric structure steps up from the bottom electrode to a top surface of the dielectric structure.

11. The memory device according to claim 10, wherein the memory cell further comprises:

a cap layer between and contacting the top electrode and the dielectric structure, wherein the cap layer is conductive and depends upon less energy to react with oxygen than the top electrode.

12. The memory device according to claim 10, further comprising:

a conductive filament in the dielectric structure, wherein the conductive filament extends from the top surface of the dielectric structure towards a bottom surface of the dielectric structure and terminates before the bottom surface of the dielectric structure.

13. An integrated circuit (IC) chip comprising a memory cell, wherein the memory cell comprises:

a first electrode;

a second electrode overlying the first electrode; and

a plurality of dielectric layers stacked between the first and second electrodes,

wherein the plurality of dielectric layers comprise a first dielectric layer, a second dielectric layer overlying the

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first dielectric layer, and a third dielectric layer overlying the second dielectric layer, the first dielectric layer is closest to the first electrode amongst the plurality of dielectric layers, the second dielectric layer has a bandgap larger than a bandgap of the first dielectric layer, and the third dielectric layer has a bandgap smaller than the bandgap of the second dielectric layer and larger than the bandgap of the first dielectric layer.

14. The IC chip according to claim 13, wherein a top valence band edge of the first dielectric layer is highest amongst the plurality of dielectric layers.

15. The IC chip according to claim 13, wherein a bottom conductive band edge of the first dielectric layer is closest to a fermi level of the first electrode amongst the plurality of dielectric layers.

16. The IC chip according to claim 13, further comprising: a metal cap layer between and directly contacting the second electrode and a topmost one of the plurality of dielectric layers, wherein the metal cap layer has a higher affinity for oxygen than the second electrode.

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17. The IC chip according to claim 15, further comprising: a substrate; a transistor over the substrate; and an interconnect structure over and electrically coupled to the transistor, wherein the memory cell is in the interconnect structure, and wherein the first electrode is electrically shorted to a source/drain region of the transistor by the interconnect structure.

18. The memory device according to claim 10, wherein a top valence band edge of the dielectric structure steps down from the bottom electrode to the top surface of the dielectric structure.

19. The IC chip according to claim 13, wherein the third dielectric layer has an electron affinity between that of the first dielectric layer and that of the second dielectric layer.

20. The IC chip according to claim 14, wherein a top valence band edge of the third dielectric layer is between a top valence band edge of the second dielectric layer and the top valence band edge of the first dielectric layer.

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